Rigidity Theory

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0. A brief history

Origin

The study of the rigidity of frameworks can be traced back to a claim made by **Euler** in 1776 that "A closed spatial figure allows not changes, as long as it is not ripped apart."

Cauchy, in 1813, first established a foundation related to Euler's conjecture by proving Cauchy's rigidity theorem that every triangulated convex spherical surface in $\mathbb{R}^3$ is rigid.

> **Cauchy's rigidity theorem**
>
> If two 3-dimensional convex poly hedra P and P' are combinatorially equivalent with corresponding facets being congruent, then also the angles between corresponding pairs of adjacent facets are equal (and thus P is congruent to P')

After about 200 years, **Alexandrov(1940's)** slightly generalize Cauchy's rigidity theorem and **Connelly(1977)** disproved such an Euler's claim by providing a counterexample about flexible polyhedra.

Combinatorial and Algebraic characterizations

Rigidity of frameworks is often analyzed **combinatorially** through the notion of **'generic'** framework.

In 1864, **Maxwell** introduced what is now known as Maxwell's count, a necessary condition for the minimum number of edges required to make 1, 2 and 3-dimensional framework with n vertices rigid.

> **Maxwell's Count**
>
> If a 1-dimensional framework is rigid, then $e(G) \geq n(G) - 1$.
> If a 2-dimensional framework is rigid, then $e(G) \geq 2n(G) - 3$.
> If a 3-dimensional framework is rigid, then $e(G) \geq 3n(G) - 6$.

This condition is further extended to a d-dimensional framework ($e(G) \geq dn(G) - \binom{d+1}{2}$).

After about 100 years, in 1970, **Laman** completely characterizes 2-dimensional rigidity by proving combinatorial necessary and sufficient condition for a 2-dimensional framework to be rigid by extending Maxwell's count.

While combinatorial conditions were successfully established for 2D generic frameworks, directly extending them to 3D or higher dimension is not possible due to counterexamples such as the 'double banana' framework.

Instead, **Asimow and Roth(1978, 1979)** pioneered an **algebraic approach**.

They formally defined the Rigidity Matrix as the Jacobian of the distance function and proved that for generic configurations, infinitesimal rigidity is equivalent to rigidity.

However, the assumption of 'generic' positions is crucial. **Saxe (1979)** proved that determining whether a specific (non-generic) framework can be embedded in a given dimension is NP-Hard. This finding theoretically justified the necessity of the generic assumption in rigidity analysis.

Global rigidity and Modern Developments

Beyond (local) rigidity, study has expanded to **Global rigidity**, which asks the uniqueness of realization.

Hendrickson(1992) proposed combinatorial necessary conditions for a d-dimensional generic framework to be globally rigid, which were later proven to be sufficient in $\mathbb{R}^2$ by **Jackson and Jordán(2005).**

For higher dimensions, **Connelly(2005)** proved a sufficient conditions for a d-dimensional generic framework to be globally rigid by introducing 'stress matrix'.
Gortler(2010) proved that such a sufficient conditions proposed by Connelly is necessary as well and hence established a complete algebraic condition for generic global rigidity.

More recently, **Tanigawa(2015)** provided a bridge between combinatorial and global rigidity by proving that vertex-redundant rigidity is a sufficient condition for generic global rigidity in any dimension.

However, obtaining a complete combinatorial characterization for d≥3 remains a major open problem. In 2026, **Baranyai** proposed an exact combinatorial characterization of 3-dimensional generic rigidity, but **Dewar(2026)** found a counterexample of sufficiency of this characterization.

In short, while d-dimensional generic globally rigid framework is completely characterized algebraically, a complete combinatorial characterization for $d \geq 3$ has yet not been found.

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1. Preliminaries

1.1. Basic definitions

Def 1.1) Framework

> A ***framework*** in $\mathbb{R}^d$ is a pair (G, p), where G is a finite simple graph and $p : V(G) \to \mathbb{R}^d$ is a (not necessarily injective) map. We also say that p is a d-dimensional ***realization*** of G. Sometimes, we denote $f_p = (G, p)$.

Def 1.2) Joint, Bar

> For $v_i \in V(G)$, we say that $p_i := p(v_i)$ is the ***joint*** of (G, p) corresponding to v_i.
> Also, for $v_iv_j \in E(G)$, we say that $\{p_i, p_j\}$ is the ***bar*** of (G, p) corresponding to v_iv_j.
> The ***length of the bar*** $\{p_i, p_j\}$ is defined as $L(\{p_i, p_j\}) := \|p(v_i) - p(v_j)\|_2$.

Def 1.3) Motion, rigid and flexible framework

A ***(finite) motion*** of (G, p) is an indexed family $\{P_i\}_{i\in[n]}$of functions $P_i : [0,1] \to \mathbb{R}^d$ s.t.

1. (Initial Condition) $P_i(0) = p(v_i), \quad \forall i \in [n]$.
2. (Differentiability) $P_i(t)$ is differentiable on $[0,1], \quad \forall i \in [n]$.
3. (Distance-preserving) $\|P_i(t) - P_j(t)\| = \|p_i - p_j\|$ for all $t \in [0,1]$ and $v_iv_j \in E(G)$.

A motion $\{P_i\}$ is called ***rigid*** if $\|P_i(t) - P_j(t)\| = \|p_i - p_j\|$ for all $t \in [0,1]$ and $(i,j) \in [n]^2$.
Also, $\{P_i\}$ is called ***flexible*** if $\exists (i,j) \in [n]^2 \setminus E(G)$ s.t. $\|P_i(t) - P_j(t)\| \neq \|p_i - p_j\|$ for all $t \in (0,1]$.

With these definitions, (G, p) is called rigid if there is no flexible motion. Otherwise, (G, p) is called flexible.

Remark

1. The term **'differentiable'** can be replaced by the term **'continuous'**.
2. The term **'for all** $t \in (0,1]$**'** in the definition of flexible motion can be replaced by the term **'for some** $t \in (0,1]$**'.**

Following **Klein's approach** in geometry, the set of all rigid motion with composition is a **Euclidean group** $\mathrm{E}(d)$. So, applying a rigid motion is an **action of a group** $\mathrm{E}(d)$ **on** $\mathbb{R}^{dn}$**.**

1.2. Equivalence & Congruence

So far, we defined a framework with a tuple of a simple graph and a realization. However, for a formation system(engineering aspect), the set of edges $E(G)$ could be defined by a set of constraints. Here, the constraint means a task or a goal that agents work together to achieve. (e.g. communication network, distance-preserving, ...)

So, we can alternatively describe a formation by $f_p = (\mathbf{V}, \mathbf{C}, p)$ where $\mathbf{V} = [n]$ and $\mathbf{C}$ be a set of constraints. Obviously, $|\mathbf{C}| = e(G)$.

In this aspect, we can introduce more definitions.

Def 1.4) Equivalence / Congruence of frameworks

Two frameworks $f_p = (G, p) = (\mathbf{V}, \mathbf{C}, p)$ and $f_q = (\mathbf{V}, \mathbf{C}, q) = (G, q)$ with different realizations p, q, are called...

- **Equivalent** if $\mathrm{mag}[c_i(p(x_i))] = \mathrm{mag}[c_i(q(x_i))]$ for all $c_i \in \mathbf{C}$ and denoted by $f_p \sim f_q$ where $x_i \subset \mathbf{V}$ is the set of nodes involved in the ith constraint c_i. In this context, $p(x_i)$ and $q(x_i)$ represent the realized points(coordinates) of these nodes in each framework, and $\mathrm{mag}[\cdot]$ is the exact value of the argument.
- **Congruent** if $\mathrm{mag}[c_i(p(x_i))] = \mathrm{mag}(c_i(q(x_i)))$ for all $c_i \in \mathbf{C}_K$ and denoted by $f_p \equiv f_q$ where $\mathbf{C}_K$ is the complete constraint set containing constraints for every possible pair of vertices in the graph G. This corresponds to the set of all edges in the complete graph K_n. In this case, x_i represents the set of nodes involved in each constraint c_i within the expanded set $\mathbf{C}_K$.

For example, if the constraint is given as a distance between each vertices, then $c_i \in \mathbf{C}$ is the distance constraint on the ith edge and x_i is the set of endpoints of the ith edge.

So, if $uv \in E(G)$ is the ith edge, then $x_i = \{u, v\}$ and $c_i(p(x_i)) = \|p(u) - p(v)\|$. In this sense, we can describe the equivalence and congruence for distance constraint only.

Two frameworks f_p and f_q are called...

- **Distance equivalent** if $\|p_i - p_j\| = \|q_i - q_j\|$ for all $v_i v_j \in E(G)$
- **Distance congruent** if $\|p_i - p_j\| = \|q_i - q_j\|$ for all $(i, j) \in [n]^2$.

Def 1.5) Constraint function

> For a framework $f_p = (G, p) = (\mathbf{V}, \mathbf{C}, p)$, the constraint function $h_G : \mathbb{R}^{dn} \to \mathbb{R}^m$ $(m = e(G))$ is defined by
>
> $$h_G(p) := \begin{bmatrix} \text{mag}[c_1(p(x_1))] \\ \vdots \\ \text{mag}[c_m(p(x_m))] \end{bmatrix}$$
>
> for square of distance constraint,
>
> $$h_G(p) = \begin{bmatrix} \text{mag}[c_1(p(x_1))] \\ \vdots \\ \text{mag}[c_m(p(x_m))] \end{bmatrix} = \begin{bmatrix} \|p_{j_1} - p_{i_1}\|^2 \\ \vdots \\ \|p_{j_m} - p_{i_m}\|^2 \end{bmatrix}$$
>
> and it is called an 'edge function'.

Formally, edge function can be defined as:

Def 1.5.1) Edge function

> Let (G, p) be a framework with graph G and realization p. Then, the ***edge function*** $g_G : \mathbb{R}^{dn} \to \mathbb{R}^m$ where $m = e(G)$ is defined by
>
> $$g_G(p) = \frac{1}{2}\left[\|p_i - p_j\|^2\right]_{(i,j) \in E(G)}$$

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2. Rigidity and infinitesimal rigidity

2.1. Basic rigidity concepts

Q. What is the "Rigid framework" ?

Def 2.1) Rigidity (Alternative definition)

> A framework $f_p = (G, p)$ is called ***rigid*** in a given Euclidean space $\mathbb{R}^d$ if there exists a nbd. U_p of p s.t. every framework $f_q = (G, q)$ with $q \in U_p$ satisfy $f_p \sim f_q \Rightarrow f_p \equiv f_q$.

It can be described in terms of constraint function function like below.

$f_p = (G, p)$ is rigid in $\mathbb{R}^d$ if there exists a nbd. U_p of p s.t.

$$h_G^{-1}(h_G(p)) \cap U_p = h_K^{-1}(h_K(p)) \cap U_p \quad \cdots \quad (*)$$

Here,

- LHS = the set of frameworks in U_p satisfying the constraint $\mathbf{C}$.
- RHS = the set of frameworks in U_p satisfying the constraint $\mathbf{C}_K$.

Obviously, there is only one congruent framework satisfying $\mathbf{C}_K$. Also, since $\mathbf{C} \subset \mathbf{C}_K$, $\mathrm{LHS} \supset \mathrm{RHS}$. So, this expression means that, if $\mathrm{LHS} \neq \mathrm{RHS}$, then there is at least one framework that is distinct (non-congruent) from that of in RHS and hence f_p is not rigid.

Def 2.2) Distance rigidity

> A framework f_p is called ***distance rigid*** in $\mathbb{R}^d$ if there exists $\varepsilon > 0$ s.t. for any framework f_q that is locally perturbed from f_p (i.e. $q_i \in B_\varepsilon(p_i),\ \forall i$), f_p and f_q are distance equivalent only if f_p and f_q are distance congruent.

This means that f_p is distance rigid if every framework satisfying same edge-distance constraints to f_p near p is congruent to f_p (Locally unique embedding satisfying edge-distance constraint).

As we've observed above, the rigidity can be described algebraically like $(*)$. Introduce a definition for better understanding.

Def 2.3) Configuration space

> Given a framework (G, p) in $\mathbb{R}^d$ with $n(G) = n$, a set of realizations
>
> $$\mathcal{C}(G, p) = \{q \in \mathbb{R}^{dn} : \|q_i - q_j\|^2 = \|p_i - p_j\|^2 \text{ for all } ij \in E(G)\}$$
>
> is called a ***configuration space***, and it can be written as an inverse image of edge function f_G
>
> $$\mathcal{C}(G, p) = f_G^{-1}(f_G(p)).$$
>
> Intuitively, $\mathcal{C}(G, p)$ is the set of realizations q that makes $(G, q) \sim (G, p)$.

Recall that applying rigid motions to a framework (G, p) is an action of a group $\mathrm{E}(d)$ on $\mathbb{R}^{dn}$. Then, the set of all realizations q that makes (G, q) congruent to (G, p) is an orbit of $p \in \mathbb{R}^{dn}$ under $\mathrm{E}(d)$, i.e.

In the view of group action

$$\mathcal{O}_p = \mathrm{orb}_{\mathrm{E}(d)}(p) = \{s(p) : g \in \mathrm{E}(d)\} = \text{realizations } q \text{ that makes } (G, q) \equiv (G, p)$$

Now, obviously, $\mathrm{orb}_{\mathrm{E}(d)}(p) = \mathcal{C}(K_n, p) \subset \mathcal{C}(G, p)$ if $n(G) = n$.

So, we can rewrite $(*)$ by

$$\mathcal{C}(G, p) \cap U_p = \mathcal{C}(K_n, p) \cap U_p = \mathrm{orb}_{\mathrm{E}(d)}(p) \cap U_p$$

only for distance rigidity.

Q. Are there two definition of distance rigidity? $\rightarrow$ No.

Thm 2.4) Equivalent of definitions of rigidity

> Two definitions of distance rigidity (in the view of motion and set-theoretic definition) of framework is equivalent.

Proof

Define two sets first.

$$M_p := \{q \in \mathbb{R}^{dn} : \|q_i - q_j\| = \|p_i - p_j\|,\ \forall v_i v_j \in E(G)\}$$
$$H_p := \{q \in \mathbb{R}^{dn} : \|q_i - q_j\| = \|p_i - p_j\|,\ \forall (i,j) \text{ s.t. } 1 \le i < j \le n\}$$

Here, M_p and H_p denotes the set of realizations equivalent to p and congruent to p, respectively. Also, obviously, $H_p \subset M_p$.

($\Rightarrow$) Suppose that a framework (G,p) is flexible by the definition 1.3 (i.e. in the view of motion). Then, $\exists$ a flexible motion $\{P_i\}_{i\in[n]}$, i.e.

$$\exists (i,j) \in [n]^2 \setminus E(G) \text{ s.t. } \|P_i(t) - P_j(t)\| \neq \|p_i - p_j\|, \quad \text{for some } t \in (0,1]$$

In other words, there exists a function $P : (0,1] \to \mathbb{R}^{dn}$ defined by

$$P(t) = [P_1(t) \quad P_2(t) \quad \cdots \quad P_n(t)]^\mathsf{T}$$

s.t. $P(0) = p$, $P(t) \in M_p$ for all t and $P(t) \notin H_p$ for some t.

So, for some t, for any $\varepsilon > 0$, we can find a time $\delta \in (0,1]$ small enough s.t. $\|P(\delta) - p\| < \varepsilon$ and $P(\delta) \notin H_p$.

Thus, $q := P(\delta)$ satisfies

$$q \in B_\varepsilon(p) \cap (M_p \setminus H_p)$$

and hence (G,p) is flexible by the definition 2.2 (set-theoretic).

($\Leftarrow$) Conversely, suppose that a framework (G,p) is not rigid by definition 2.2.

Claim 1: There exist infinitely many realizations that are equivalent but not congruent to p near p.

Given $\varepsilon > 0$,

$$\exists q \in B_\varepsilon(p) \cap (M_p \setminus H_p)$$

Suppose NOT: There exists only finitely many realizations $q_1, \cdots, q_N$ that are equivalent but not congruent to p near p. WLOG, let $d(p, q_m) = \varepsilon_m$ and $\varepsilon_1 > \cdots > \varepsilon_N$.

Then, by choosing ε smaller than ε_N, every realizations in $B_\varepsilon(p)$ equivalent to p is congruent to p as well and hence contradicts the assumption that (G,p) is flexible.

So, we have a collection of infinitely many number of realizations $\{q_\alpha\}_{\alpha\in J}$(where J is an index set) that are equivalent but not congruent to p, i.e. $q_\alpha \in M_p \setminus H_p$ for all $\alpha \in J$.

Moreover, we can take a countably infinite subsequence $\{q_k\}$ of $\{q_\alpha\}$ corresponding to a sequence of distance $\varepsilon_1 > \varepsilon_2 > \cdots$.

Then, $q_k \to p$ as $k \to \infty$ and hence we can conclude that p is a limit point of $M_p \setminus H_p$.

Claim 2: There exists an analytic curve passing through p into $M_p \setminus H_p$.

Since M_p is an algebraic variety and H_p is its subvariety(defined by polynomial equations), $M_p \setminus H_p$ is a semi-algebraic set.

Since p belongs to $\overline{M_p \setminus H_p}$ from claim 1, by the The Curve Selection Lemma, there exists a real analytic curve $P : [0,1) \to \mathbb{R}^{dn}$ s.t.

$$P(0) = p \quad \text{and} \quad P(t) \in M_p \setminus H_p$$

This curve $P(t)$ represents a finite flex of (G,p) and hence it is flexible by definition 1.3. $\square$

Def 2.5) Global distance rigidity

> A framework (G,p) is called ***globally distance rigid*** in $\mathbb{R}^d$ if "every" framework (G,q) equivalent to (G,p) is also congruent to (G,p) in $\mathbb{R}^d$.

Henceforth, unless otherwise specified, the term 'rigidity' will refer exclusively to **'distance rigidity'.**

2.2. Infinitesimal rigidity

Motivation: It is in general very difficult to determine whether a given framework is rigid or not since it requires solving a system of quadratic equations. It is therefore common to **linearize** the problem by differentiating the equations.

Let $\{P_i\}$ be a finite motion of a 2-dimensional framework (G,p) and let $\{p_i, p_j\}$ be a bar of length l. Then,

$$\|P_i(t) - P_j(t)\|^2 = \left[P_i^{(1)}(t) - P_j^{(1)}(t)\right]^2 + \left[P_i^{(2)}(t) - P_j^{(2)}(t)\right]^2 = l^2 \quad \forall t \in [0,1]$$

where $P_i^{(k)}(t)$ is the kth coordinate of p_i at t.

By differentiating both sides w.r.t. t,

$$\left(P_1^{(1)}(t) - P_j^{(1)}(t), P_1^{(2)}(t) - P_j^{(2)}(t)\right) \cdot \left(\left[P_1^{(1)}(t) - P_j^{(1)}(t)\right]', \left[P_1^{(2)}(t) - P_j^{(2)}(t)\right]'\right) = 0$$

Evaluating at $t = 0$, we obtain

$$\left(p_i^{(1)} - p_j^{(1)}, p_i^{(2)} - p_j^{(2)}\right) \cdot \left(u_i^{(1)} - u_j^{(1)}, u_i^{(2)} - u_j^{(2)}\right) = 0$$

or in short $(p_i - p_j) \cdot (u_i - u_j)$. Here, u_i, u_j are initial velocity vectors of p_i and p_j, respectively. So, this implies that the difference between initial velocity vectors is orthogonal to the line through p_i and p_j.

These vectors preserve the length of $\{p_i, p_j\}$ at first order.

Def 2.6) Constricted framework

> A d-dimensional framework (G,p) is called ***constricted*** if all points in $\mathcal{P} = \{p_1, \cdots, p_n\}$ lie in $k \geq d-1$ dimensional affine subspace. Otherwise, (G,p) is called ***normal***.

Thus, a framework is constricted if it is geometrically degenerated.

Def 2.7) Infinitesimal motion / Infinitesimal rigidity

> An ***infinitesimal motion*** of a framework (G, p) is a function $u : V(G) \to \mathbb{R}^d$ s.t.
>
> $$(p_i - p_j) \cdot (u_i - u_j) = 0 \quad \forall v_i v_j \in E(G)$$
>
> That is, for a distance constrained graph G, a function consists of dn instantaneous velocity vectors of the nodes that do not change the equivalence conditions between nodes.
>
> Here, an infinitesimal motion is called ***rigid*** if
>
> $$(p_i - p_j) \cdot (u_i - u_j) = 0 \quad \forall (i, j) \in [n]^2$$
>
> Also, an infinitesimal motion is called ***flex*** if
>
> $$(p_i - p_j) \cdot (u_i - u_j) \neq 0 \quad \exists v_i v_j \notin E(G)$$
>
> In this sense, (G, p) is called ***infinitesimally rigid*** if every infinitesimal motion of (G, p) is an infinitesimal rigid motion. Otherwise, (G, p) is said to be ***infinitesimaly flexible***.

Note: In some context, infinitesimal rigid motion also can be called by a "Trivial motion" or a "Special Euclidean motion".

- **Trivial motions** is an infinitesimal motions include only the translational and rotational motions. (infinitesimal rigid motion)
- **Special Euclidean motion** is defined by the motions of nodes that keep the congruence conditions among the nodes. It contains the translations and rotations, and hence Special Euclidean motion is equivalent to the **direct isometries.**

In short,

> **Equivalence of definitions**
>
> - Infinitesimal rigid motion = Trivial motion
> - Special Euclidean motion = Direct isometry

However, **in the infinitesimal viewpoint, the four of them are all equivalent.**

Remark: All constricted frameworks infinitesimally flexible.

- Now, we can come up with a very natural question.

Q. Is there any relationship between the rigidity and the infinitesimal rigidity?

Thm 2.8) Infinitesimal rigidity is stronger than rigidity

> If a framework f_p is infinitesimally rigid, then f_p is rigid.

- This theorem enables for us to solve the first-order equations(infinitesimal rigidity) instead of solving the second-order equation(rigidity).
- We will prove this theorem after introducing the **rigidity matrix**, which is a powerful tool for analyzing infinitesimal rigidity.

2.2.1. Rigidity Matrix

In section 1, we've introduced a constraint function and noted that it is called an **'edge function'** in some context if its value is square of length of each edge.

Begin with formally restating the definition of edge function and introducing an important concept directly derived from an edge function.

Def 2.9) Rigidity Matrix

> Let (G, p) be a framework with graph G and realization p. Then, the ***edge function*** $g_G : \mathbb{R}^{dn} \to \mathbb{R}^m$ where $m = e(G)$ is defined by
>
> $$g_G(p) = \frac{1}{2}\left[\|p_i - p_j\|^2\right]_{(i,j)\in E(G)}$$
>
> Then, the ***rigidity matrix*** is defined by the Jacobian of rigidity matrix.
>
> $$\mathbf{R}(G, p) = \frac{\partial g_G(p)}{\partial p} \in \mathbb{R}^{m\times dn}$$
>
> Here, each row and column corresponds to each edges and nodes of the framework, respectively.

Explicitly, the rigidity matrix is expressed as

$$\mathbf{R}(G,p) = \begin{pmatrix} & & & & & \vdots & & & & & \\ 0 & \dots & 0 & p_i - p_j & 0 & \dots & 0 & p_j - p_i & 0 & \dots & 0 \\ & & & & & \vdots & & & & & \end{pmatrix}$$

Let $e_k = v_i v_j \in E(G)$ is the kth edge with endpoints v_i and v_j. Then, it occupies the $d(i-1)+1^{\text{th}}$ to di^{th} columns and $d(j-1)+1^{\text{th}}$ to dj^{th} columns of the kth row.

Here, $\left(p_i^{(1)} - p_j^{(1)}\right), \cdots, \left(p_i^{(d)} - p_j^{(d)}\right)$ are in the $d(i-1)+1^{\text{th}}$ to di^{th} columns of the kth row.

Similarly, $\left(p_j^{(1)} - p_i^{(1)}\right), \cdots, \left(p_j^{(d)} - p_i^{(d)}\right)$ are in the $d(j-1)+1^{\text{th}}$ to dj^{th} columns of the kth row.

Example (Tetrahedron)

For a formation (G, p) of regular tetrahedron in $\mathbb{R}^3$,
$V(G) = \{v_1, v_2, v_3, v_4\}$ and $E(G) = \{v_1v_2, v_1v_3, v_1v_4, v_2v_3, v_2v_4, v_3v_4\}$ and
$p = \left[p_1^\top\ p_2^\top\ p_3^\top\ p_4^\top\right]^\top$ where $p_i = [x_i\ y_i\ z_i]$.

Note that $|V(G)| = n(G) = 4,\ |E(G)| = e(G) = 6,\ d = 3.$

Then, by definition, the rigidity matrix be

$$\mathbf{R}(p) = \begin{bmatrix} (p_1-p_2)^\top & (p_2-p_1)^\top & 0^\top & 0^\top \\ (p_1-p_3)^\top & 0^\top & (p_3-p_1)^\top & 0^\top \\ (p_1-p_4)^\top & 0^\top & 0^\top & (p_4-p_1)^\top \\ 0^\top & (p_2-p_3)^\top & (p_3-p_2)^\top & 0^\top \\ 0^\top & (p_2-p_4)^\top & 0^\top & (p_4-p_2)^\top \\ 0^\top & 0^\top & (p_3-p_4)^\top & (p_4-p_3)^\top \end{bmatrix} \in \mathbb{R}^{6\times(3\times 4)}$$

Let $p_1 = (1,1,1),\ p_2 = (1,-1,-1),\ p_3 = (-1,1,-1),\ p_4 = (-1,-1,1)$ be coordinates of each vertex. Then, the rigidity matrix be

$$\mathbf{R}(p) = \left[\begin{array}{ccc|ccc|ccc|ccc} 0 & 2 & 2 & 0 & -2 & -2 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 2 & 0 & 2 & 0 & 0 & 0 & -2 & 0 & -2 & 0 & 0 & 0 \\ \hline 2 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2 & -2 & 0 \\ \hline 0 & 0 & 0 & 2 & -2 & 0 & -2 & 2 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 2 & 0 & -2 & 0 & 0 & 0 & -2 & 0 & 2 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & -2 & 0 & -2 & 2 \end{array}\right]$$

Note that infinitesimal motion is a function of velocity vectors of each nodes. So, there are d velocity components for each nodes and hence one infinitesimal motion can be identified as a dn-dimensional column vector $\mathbf{u}$.

- Recall: $u : V(G) \to \mathbb{R}^d$ is an infinitesimal motion if $(p_i - p_j) \cdot (u_i - u_j) = 0$ for all $v_i v_j \in E(G)$.

From here, we can find some useful property.

Prop 2.10) Infinitesimal motion with rigidity matrix

> The set of infinitesimal motion of framework (G,p) is the kernel of $\mathbf{R}(G,p)$.

Proof

Proof) Let $\mathcal{M}(G,p)$ be the set of infinitesimal motion of (G,p) and $\mathbf{u} \in \mathcal{M}(G,p)$.

Let $e_k = v_i v_j$. Then, the kth row of $\mathbf{R}(G,p)\mathbf{u}$ be

$$[\mathbf{R}(G,p)\mathbf{u}]_k = (p_i - p_j) \cdot u_i + (p_j - p_i) \cdot u_j = (p_i - p_j) \cdot (u_i - u_j)$$

Here, since $\mathbf{u}$ is a column vector representation of an infinitesimal motion, $\mathrm{RHS} = 0$ and hence $\mathbf{u} \in \mathrm{Ker}(R(G,p))$.

Thus, $\mathcal{M}(G,p) \subset \mathrm{Ker}(R(G,p))$.

Conversely, given $\mathbf{x} \in \mathrm{Ker}(\mathbf{R}(G,p))$, $[\mathbf{R}(G,p)\mathbf{x}]_k = 0$ for each k and hence $(p_i - p_j) \cdot (x_i - x_j) = 0$.

Thus, $\mathbf{x}$ be an infinitesimal motion of (G,p) and hence $\mathrm{Ker}(\mathbf{R}(G,p)) \subset \mathcal{M}(G,p)$.

$\therefore \mathcal{M}(G,p) = \mathrm{Ker}(\mathbf{R}(G,p))$. $\square$

From this fact, by letting $\mathcal{R}(G,p)$ be the set of all infinitesimal rigid motion,

$$\mathcal{R}(G,p) \subset \mathcal{M}(G,p)$$

by definition.

Also, (G,p) is infinitesimally rigid if and only if $\mathcal{R}(G,p) = \mathcal{M}(G,p)$.

By the previous proposition, the following fact directly follows.

Lem 2.11) Dimension and infinitesimal rigidity

> Let (G, p) be a normal framework in $\mathbb{R}^d$. Then,
>
> $$\dim \mathcal{R}(G,p) \leq \dim \mathcal{M}(G,p) = \text{nullity}(\mathbf{R}(G,p))$$
>
> Equivalently, by rank-nullity theorem,
>
> $$\text{rank}(\mathbf{R}(G,p)) \leq dn - \dim \mathcal{R}(G,p)$$
>
> The equality holds if and only if (G, p) is infinitesimally rigid.

Here, while we know d, n but we don't know about $\dim \mathcal{R}(G,p)$ and hence we need to find it.

To find it, we assume **2-dimensional general framework** for the following arguments. They can be directly generalized to an arbitrary dimension.

Def 2.12) General Position

> $\mathcal{P} = \{v_1, \cdots, v_n\} \subseteq \mathbb{R}^d$ is in ***general position*** if for $1 \leq m \leq d$, no $m+1$ points in $\mathcal{P}$ lie in an $(m-1)$-dimensional affine subspace of $\mathbb{R}^d$. A framework whose joints are in general position is called a ***general framework***.

The following definition explains about two types of infinitesimal motions.

Def 2.13) Infinitesimal translation / Infinitesimal rotation ($\mathbb{R}^2$ ver.)

> An ***infinitesimal translation of the plane*** is of the form
>
> $$\tau_{(u,v)} : \mathbb{R}^2 \to \mathbb{R}^2 \; ; \quad \tau_{(u,v)}(x,y) = (u,v)$$
>
> This implies that every point (x, y) in $\mathbb{R}^2$ moves with same velocity (u, v).
>
> An ***infinitesimal rotation of the plane*** is of the form
>
> $$a\rho_{(x_0,y_0)} : \mathbb{R}^2 \to \mathbb{R}^2 \; ; \quad a\rho_{(x_0,y_0)}(x,y) = a(y_0 - y, x - x_0)$$

Let r be a radius vector from a center (x_0, y_0) to (x, y), i.e. $r = (x - x_0, y - y_0)$.
Then, the rotation vector v of (x, y) must satisfy $r \perp v$ and hence $r \cdot v = 0$. So, v must be of the form $a(-(y - y_0), x - x_0) = a(y_0 - y, x - x_0)$ where $a \in \mathbb{R}$.

Lem 2.14) Dimension of infinitesimal translations and rotations

> The set of all infinitesimal translations and rotations of $\mathbb{R}^2$ forms a 3-dimensional vector space.

Proof

Clearly, $\tau_{(1,0)}, \tau_{(0,1)}$ and $\rho_{(0,0)}$ are independent infinitesimal motions of the plane.

Claim: For any $a, b, c \in \mathbb{R}$,

$$a\rho_{(0,0)} + b\tau_{(1,0)} + c\tau_{(0,1)} = \begin{cases} \tau_{(b,c)} & \text{if } a = 0 \\ a\rho_{(-c/a, b/a)} & \text{if } a \neq 0 \end{cases}$$

First, if $a = 0$, then $b\tau_{(1,0)} + c\tau_{(0,1)} = b(1,0) + c(0,1) = (b,0) + (0,c) = (b,c) = \tau_{(b,c)}$.
Next, if $a \neq 0$, then

$$\begin{aligned}\big(a\rho_{(0,0)} + b\tau_{(1,0)} + c\tau_{(0,1)}\big)(x,y) &= \big(a\rho_{(0,0)} + \tau_{(b,c)}\big)(x,y) \\ &= a\rho_{(0,0)}(x,y) + \tau_{(b,c)}(x,y) \\ &= a(-y,x) + (b,c) \\ &= (b - ay, c + ax) = a\rho_{(-c/a,b/a)}(x,y)\end{aligned}$$

This implies that given an infinitesimal translation $\tau_{(b,c)}$ can be expressed by a linear combination of translational basis elements $(\tau_{(1,0)}, \tau_{(0,1)})$, and given an infinitesimal rotation $a\rho_{(x_0,y_0)}$ can be expressed by a linear combination of three basis elements. □

Q. What about for d-dimensional space?

From the proof of this lemma, we can observe that the dimension of the space of all infinitesimal translations and rotations are **the sum of the number of independent infinitesimal translations and rotations.**

$$\text{Dimension} = (\#\text{ independent inf.translations }) + (\#\text{ independent inf.rotations })$$

For d-dimensional space, obviously, **the number of independent infinitesimal translations is** d. However, **the number of independent infinitesimal rotations is** $\binom{d}{2}$ because each rotation is defined on a 'plane', which is determined by two independent vectors.

Thus, the set of all infinitesimal translations and rotations of $\mathbb{R}^d$ forms a $d + \binom{d}{2} = \binom{d+1}{2}$-dimensional vector space.

Now, recall that for infinitesimal viewpoint, <u>infinitesimal rigid motion is equal to direct isometry</u> and hence **infinitesimal rigid motion consists of infinitesimal translation and rotation only.**

The following lemma describes about this fact.

Lem 2.15) Classifications fo infinitesimal rigid motions

> Let (G,p) be a 2-dimensional general framework with $V(G) = \{v_1, \cdots, v_n\}$, $n \geq 2$. Then $\mathcal{R}(G,p)$ is the 3-dimensional space of infinitesimal translations and rotations restricted to the points $\mathcal{P} = \{v_1, \cdots, v_n\}$ of (G,p).

Proof

The proof consists of two steps.

For a restriction map $(\cdot)|_{\mathcal{P}} : I'_d(\mathbb{R}^2) \to \mathcal{R}(G,p);\quad f \mapsto f|_{\mathcal{P}}$, (where $I'_d(\mathbb{R}^2)$ is a set of infinitesimal direct isometries, i.e. infinitesimal translations and rotations)

- **Surjectivity:** Given $u \in \mathcal{R}(G,p)$, $\exists \bar{u} \in I'_d(\mathbb{R}^2)$ s.t. $\bar{u}|_{\mathcal{P}} = u$.
- **Injectivity:** Given $\kappa, \lambda \in I'_d(\mathbb{R}^2)$ and corresponding restriction $\kappa|_{\mathcal{P}}, \lambda|_{\mathcal{P}} \in \mathcal{R}(G,p)$, $\kappa|_{\mathcal{P}} = \lambda|_{\mathcal{P}} \Rightarrow \kappa = \lambda$.

Surjectivity

Let $u = (u_1, \cdots, u_n) \in \mathcal{R}(G,p)$ with $u_i = (u_i^{(1)}, u_i^{(2)})$ for all $i = 1, \cdots, n$.
Clearly, the restriction of τ_{u_1} to $\mathcal{P}$ is an infinitesimal rigid motion of (G,p).

Hence $\hat{u} = u - \tau_{u_1}|_{\mathcal{P}} \in \mathcal{R}(G,p)$.

Here, note that $\hat{u} = (u_1 - u_1, u_2 - u_1, \cdots, u_n - u_1)$ and hence $\hat{u}_1 = (0,0)$. (fixing p_1)

Next, from the definition of infinitesimally rigid motion $(p_i - p_j) \cdot (u_i - u_j) = 0$ for all $(i,j) \in [n]^2$, we have

$$(p_2 - p_1) \cdot (\hat{u}_2 - \hat{u}_1) = (p_2 - p_1) \cdot \hat{u}_2 = 0$$

Since $(p_2 - p_1) \cdot \hat{u}_2 = \left(p_2^{(1)} - p_1^{(1)}, p_2^{(2)} - p_1^{(2)}\right) \cdot \hat{u}_2 = 0$,
$\hat{u}_2$ must be of the form $\hat{u}_2 = a\left(p_2^{(1)} - p_1^{(1)}, -(p_2^{(2)} - p_1^{(2)})\right)$.

So, $\hat{u}_2^{(1)} = a\left(p_2^{(1)} - p_1^{(1)}\right)$ and $\hat{u}_2^{(2)} = a\left(p_1^{(2)} - p_2^{(2)}\right)$ and hence

$$a = \frac{\hat{u}_2^{(1)}}{p_2^{(2)} - p_1^{(2)}} = \frac{\hat{u}_2^{(2)}}{p_1^{(1)} - p_2^{(1)}} \quad \text{where } p_1^{(1)} \neq p_2^{(1)} \text{ or } p_1^{(2)} \neq p_2^{(2)}$$

Now, define another motion $\bar{u} = \hat{u} - a\rho_{p_1}|_{\mathcal{P}} = u - \tau_{u_1}|_{\mathcal{P}} - a\rho_{p_1}|_{\mathcal{P}}$.
Here, since p_1 is fixed by $\hat{u}$ and the rotation $a\rho_{p_1}|_{\mathcal{P}}$ is a rotation through p_1, $\bar{u}$ fixes p_1 as well.

Also, by choosing proper rotation, we can make $\bar{u}$ fixes p_2. Then, since $\bar{u}$ fixes both p_1 and p_2, $\bar{u}_1 = \bar{u}_2 = (0,0)$.

Now, claim: **If p_1 and p_2 be fixed, then p_i are fixed for all $i > 2$.**

From the definition of infinitesimal rigid motion, we have

$$(p_i - p_1) \cdot (\bar{u}_i - \bar{u}_1) = (p_i - p_1) \cdot \bar{u}_i = 0 \quad \forall i > 2 \quad \cdots \ (*)$$
$$(p_i - p_2) \cdot (\bar{u}_i - \bar{u}_2) = (p_i - p_2) \cdot \bar{u}_i = 0 \quad \forall i > 2 \quad \cdots \ (**)$$

Since we assumed the general position, p_i is not on the line through p_1 and p_2 and hence the vectors $p_i - p_1$ and $p_i - p_2$ are not parallel.

However, since $\bar{u}_i$ is perpendicular to both of them by $(*)$ and $(**)$.

Note that in the two-dimensional plane, the only vector that is perpendicular to two vectors that are not parallel to each other is the zero vector $(0,0)$ and hence $\bar{u}_i = (0,0)$ for all $i > 2$.

Thus, $\bar{u} = u - \tau_{u_1}|_{\mathcal{P}} - a\rho_{p_1}|_{\mathcal{P}} = \mathbf{0}$ and hence $u = \tau_{u_1}|_{\mathcal{P}} + a\rho_{p_1}|_{\mathcal{P}} = (\tau_{u_1} + a\rho_{p_1})|_{\mathcal{P}}$.
This implies that any infinitesimal rigid motion u of framework (G,p) is the restriction of combination of infinitesimal translation and rotation of $\mathbb{R}^2$ to $\mathcal{P}$.

Injectivity

Suppose that κ and λ are two infintiesimal rigid motions of $\mathbb{R}^2$ s.t.

$$\kappa|_{\mathcal{P}} = \lambda|_{\mathcal{P}} = u$$

Let $x \in \mathbb{R}^2$ that is not on the line through p_1 and p_2.
Consider $\mu = \kappa - \lambda$. Then, obviously, μ is an infinitesimal rigid motion of $\mathbb{R}^2$.

Then, by the definition of infinitesimal rigid motion,

$$(\mu(x) - \mu(p_1)) \cdot (x - p_1) = 0$$
$$(\mu(x) - \mu(p_2)) \cdot (x - p_2) = 0$$

Here, since $\kappa|_{\mathcal{P}} = \lambda|_{\mathcal{P}}$, for any joint p_i, $\kappa(p_i) = \lambda(p_i)$ and hence $\mu(p_i) = (0, 0)$ for all i.
So,

$$\mu(x) \cdot (x - p_1) = 0$$
$$\mu(x) \cdot (x - p_2) = 0$$

By assumption, x is not on the line through p_1 and p_2 and hence two vectors $x - p_1$ and $x - p_2$ are not parallel. Again, since the only vector that is perpendicular to two vectors that are not parallel to each other is the zero vector $(0, 0)$, we can find that $\mu(x) = (0, 0)$.

So far, we've shown that $\mu(x) = (0, 0)$ for $x \in \mathbb{R}^2 \setminus L(p_1, p_2)$ where $L(p_1, p_2)$ is the straight line through p_1 and p_2.

Finally, let $y \in L(p_1, p_2)$. Then, obviously, y does not lie on a line through x and p_1 or x and p_2 (or both).

WLOG, let y is not on a line through x and p_1. By the definition of infinitesimal rigid motion,

$$(\mu(y) - \mu(p_1)) \cdot (y - p_1) = 0$$
$$(\mu(y) - \mu(x)) \cdot (y - x) = 0$$

and since $\mu(p_1) = \mu(x) = 0$, $\mu(y)$ is perpendicular to both $y - p_1$ and $y - x$ that are not in parallel. Thus, $\mu(y) = (0, 0)$.

Therefore, we proved that $\mu(x) = 0$ for all $x \in \mathbb{R}^2$ and hence $\kappa = \lambda$.

Hence, proved. $\square$

Intuitively, this lemma implies that the **infinitesimal rigid motions consist of translations and rotations only.** $\rightarrow$ This is why the infinitesimal rigid motion is called 'Special Euclidean Motion'.

Intuition in the view of group action

While the Euclidean group $\mathrm{E}(d)$ includes not only direct isometries(translation and rotation) but also opposite isometries(reflection and glide), the special Euclidean group $\mathrm{SE}(d)$ only includes direct isometries.

Since opposite isometries are **discontinuous jumps** and hence it is impossible to define an 'infinitesimal reflection'.

Because of this, when taking the derivative to find the infinitesimal motions (velocities), we are strictly looking at the tangent space at the identity.

Therefore, while a finite rigid motion is considered as a group action of $\mathrm{E}(d)$ to $p \in \mathbb{R}^{dn}$, the infinitesimal rigid motion is an action of Lie algebra $\mathfrak{se}(d)$ to $p \in \mathbb{R}^{dn}$.

Using previous lemma, we can obatain an important result.
Before observing it, introduce a definition first.

Def 2.16) Degree of freedom

> For a framework (G,p), a ***degree of freedom*** of (G,p) is $\text{nullity } R(G,p)$.
> Also, $\text{nullity}(\mathbf{R}(G,p)) - \dim \mathcal{R}(G,p)$ is called an ***internal degrees of freedom*** of (G,p).

Then, we can observe a degree of freedom by the following theorem.

Thm 2.17) Lower bound of DOF

> Let (G,p) be a general framework in $\mathbb{R}^d$ with $n(G) \geq d$. Then,
>
> $$\text{nullity}(\mathbf{R}(G,p)) \geq d + \binom{d}{2} = \binom{d+1}{2}$$
>
> or equivalently,
>
> $$\text{rank}(\mathbf{R}(G,p)) \leq dn - \binom{d+1}{2}$$
>
> Here, equality holds if and only if (G,p) is infinitesimally rigid.

Proof

Proof) Recall that $\dim \mathcal{R}(G,p) \leq \dim \mathcal{M}(G,p) = \text{nullity}(\mathbf{R}(G,p))$ where $\mathcal{R}(G,p)$ is the set of infinitesimally rigid motions of (G,p) and $\mathcal{M}(G,p)$ is the set of infinitesimal motions of (G,p) by lemma 2.10.

By lemma 2.14, a set of infinitesimal rigid motion consists of infinitesimal translations and rotations and hence $\dim \mathcal{R}(G,p)$ is the number of basis, i.e. the number of independent translations and rotations.

Note that there are d translations $\tau_{e_1}, \cdots, \tau_{e_d}$ where e_i is a standard basis vector and there are $\binom{d}{2}$ rotations R_{ij} where for all $i,j \in [d]$, $R_{ij} : \mathcal{P} \to \mathbb{R}^d;\ \ R_{ij}(p_k) = p_k^{(i)} e_j - p_k^{(j)} e_i$ for all k.

So, the total number of basis is $d + \binom{d}{2} = \binom{d+1}{2}$ and hence

$$\binom{d+1}{2} \leq \text{nullity}(\mathbf{R}(G,p))$$

Hence, proved. $\square$

Cor 2.18) Maximum number of edges of flexible framework

> Let (G,p) be a general framework in $\mathbb{R}^d$ with $n(G) \geq d$. If $e(G) < dn - \binom{d+1}{2}$, then (G,p) is infiinitesimally flexible.

This corollary provides the lower bound of number of edges for a framework to be rigid. **The following definition explains about a framework that has exactly this minimum threshold of edges required for rigidity.**

Now, we can prove the theorem 2.8.

Recall: Thm 2.8) Infinitesimal rigidity is stronger than rigidity

> If a framework f_p is infinitesimally rigid, then f_p is rigid.

Proof

Define a set

$$I = \{q \in \mathbb{R}^{dn} : (G, q) \text{ is infinitesimally rigid}\}.$$

Claim 1: I is an open set.

By theorem 2.16, we can rewrite such a definition by

$$I = \left\{ q \in \mathbb{R}^{dn} : \operatorname{rank}(\mathbf{R}(G, q)) = dn - \binom{d+1}{2} \right\}$$

Then, a complement of I can be described by

$$F = I^c = \left\{ q \in \mathbb{R}^{dn} : \operatorname{rank}(\mathbf{R}(G, q)) < dn - \binom{d+1}{2} \right\}$$

Recall the following fact from linear algebra.
Given a matrix M, $\operatorname{rank} M < k$ if and only if every $k \times k$ subdeterminant of M is zero.

So, F can be rewritten as

$$F = \{q \in \mathbb{R}^{dn} : \det(M_j(q)) = 0 \text{ for all } r_{\max} \times r_{\max} \text{ minors } M_j\}$$

where $r_{\max} = dn - \binom{d+1}{2}$ and $M_j(q)$ denotes the jth $r_{\max} \times r_{\max}$ submatrix of $\mathbf{R}(G, q)$.

Here, note that each component of $\mathbf{R}(G, q)$ is of the form $q_i - q_j$, which is linear and hence it is continuous w.r.t. q.

Since each subdeterminant of $\mathbf{R}(G, q)$ is a polynomial consists of sums and products of each components of $\mathbf{R}(G, q)$, the function $f_j(q) := \det M_j(q)$ is continuous w.r.t. q as well for all j.

Also, since a solution space $S = \{x : f(x) = 0\}$ for continuous function f is a closed set, F is an intersection of closed sets $F_j = \{x : f_j(q) = 0\}$ and hence F is closed, i.e.

$$F = \bigcap_j F_j \quad (\text{finite intersection of closed sets})$$

Thus, we can conclude that I is open in $\mathbb{R}^{dn}$.

Moreover, since I is open and $p \in I$ obviously, $\exists \varepsilon > 0$ s.t. $B_\varepsilon(p) \subset I$.

Now, let $q \in B_\varepsilon(p) \subset I$ is a realization close enough to p and obviously, their midpoint belongs to I as well, i.e. $\frac{p+q}{2} \in I$.

Suppose that (G, q) is equivalent to (G, p), i.e. $\|q_i - q_j\| = \|p_i - p_j\| \;\; \forall v_i v_j \in E(G)$.

Claim 2: (G, q) is congruent to (G, p).

Let $m = \frac{p+q}{2}$ is a midpoint of p and q, and consider $u = p - q$.

Then,

$$
\begin{aligned}
&(m_i - m_j) \cdot (u_i - u_j) \\
&= \left(\frac{p_i + q_i}{2} - \frac{p_j + q_j}{2} \right) \cdot ((p_i - q_i) - (p_j - q_j)) \\
&= \frac{1}{2}((p_i + q_i) - (p_j + q_j)) \cdot ((p_i - q_i) - (p_j - q_j)) \\
&= \frac{1}{2}((p_i - p_j) + (q_i - q_j)) \cdot ((p_i - p_j) - (q_i - q_j)) \\
&= \frac{1}{2}\left(\|p_i - p_j\|^2 - \|q_i - q_j\|^2 \right)
\end{aligned}
$$

Since $(G, p) \sim (G, q)$, this implies that $(m_i - m_j) \cdot (u_i - u_j) = 0$ for all $v_i v_j \in E(G)$ and hence u is an infinitesimal rigid motion of (G, m) where $m \in I$.

In other words, $u \in \ker(\mathbf{R}(G, m))$.

From lemma 2.14, each infinitesimal rigid motion is trivial motion and since (G, m) is infinitesimally rigid, each infinitesimal motion of (G, m) is infinitesimal rigid motion and hence we can conclude that u is a trivial motion.

Thus, $u = p - q$ is a trivial motion and hence we can obtain q by a sequence of translations and rotations from p.

Therefore, (G, p) is congruent to (G, q) and hence (G, p) is rigid. $\square$

Even if this proof is quite long, such a process can be summarized as:

1. I (set of realizations that makes G infinitesimally rigid) is an open set.
2. Take $q \in I$ near p.
3. Suppose that q is equivalent to p. Then, q is also congruent to p.
   Thus, each equivalent realization near p is also congruent (rigid).

Def 2.19) Minimally rigid framework

> A rigid framework (G, p) is called ***minimally rigid*** if the removal of any edges makes it flexible. Similarly, an infinitesimally rigid framework (G, p) is called ***minimally infinitesimally rigid*** if the removal of any edges makes it not infinitesimally rigid.

Def 2.20) Independent / Isostatic framework

> A framework (G, p) is ***independent*** if the rows of its rigidity matrix $\mathbf{R}(G, p)$ are linearly independent. (G, p) is ***isostatic*** if it is infinitesimally rigid and independent.

There is an intuition for independent framework.

Geometric intuition of independent framework

Claim: If (G, p) is independent, there is no redundant edge.

Setup:

Let r_i is the ith row of $\mathbf{R}(G,p)$, that is, a row corresponding to the ith edge e_i.
Given $v \in \mathbb{R}^{dn}$, $\mathbf{R}(G,p)v = 0$ implies that v is an infinitesimal motion of (G,p).

Given an edge $e_k = uw$, for its corresponding row r_k,

$$r_k \cdot v = 0$$

implies that relative velocity between vertices u and w are perpendicular to the direction of edge (i.e. no edge length change).

Here, an edge e_k is said to be **redundant** if the distance between two vertices connected by e_k is unchanged by any infinitesimal motion of framework $(G - e_k, p)$ ($d(u,w)$ is unchanged if $e_k = uw$ is removed).

Proof) Suppose that $e_k = uw$ is redundant in (G,p) and consider a framework $(G - e_k, p)$.

Then, $\mathbf{R}(G - e_k, p)$ is a rigidity matrix of $(G - e_k, p)$ and the set of infinitesimal motions of this framework is $\ker(\mathbf{R}(G - e_k, p))$.

Since e_k is redundant, given an infinitesimal motion $v \in \ker(\mathbf{R}(G - e_k, p))$, it makes the distance $d(u,w)$ unchanged, i.e. $r_k \cdot v = 0$.

Thus, we have

$$\ker(\mathbf{R}(G - e_k, p)) \subset \ker(r_k)$$

Note that if the intersection of the null spaces of linear functionals $f_1, \cdots, f_k \in V^*$ is contained in the null space of another functional $g \in V^*$, then g can be expressed by the linear combination of f_i s (reference of theorem).

So, r_k belongs to the row space of $\mathbf{R}(G - e_k, p)$ and hence

$$r_k = \sum_{i \in [m] \setminus \{k\}} c_i r_i$$

By rearranging this, we have

$$\left(\sum_{i \in [m] \setminus \{k\}} c_i r_i \right) - r_k = 0$$

and hence we can conclude that the rows of $\mathbf{R}(G,p)$ is linearly dependent.

By taking contrapositive, if (G,p) is independent, then there is no redundant edge.

This concept of 'redundancy' can be stated in the following theorem.

Thm 2.21) Equivalent conditions of isostatic framework

> For a d-dimensional general framework (G,p) with $n(G) \geq d$, TFAE:
>
> 1. (G,p) is isostatic.
> 2. (G,p) is infinitesimally rigid and $e(G) = dn - \binom{d+1}{2}$.
> 3. (G,p) is independent and $e(G) = dn - \binom{d+1}{2}$.

> 4. (G,p) is minimal infinitesimally rigid.

Briefly stating the proof:

Why $e(G) = dn - \binom{d+1}{2}$

Recall that (G,p) is infinitesimally rigid in $\mathbb{R}^d$ if and only if $\operatorname{rank}(\mathbf{R}(G,p)) = dn - \binom{d+1}{2}$.
If (G,p) is isostatic, i.e. infinitesimally rigid and independent, the rank of $\mathbf{R}(G,p)$ be

$$\operatorname{rank}(\mathbf{R}(G,p)) = m = dn - \binom{d+1}{2}$$

and hence the number of edges of isostatic framework is $dn - \binom{d+1}{2}$.

Why minimal infinitesimally rigid

As we observed, independent framework doesn't have any redundant edges and hence we cannot remove any edge to maintain the entire structure.

Def 2.22) Regular points

> Given a framework (G,p), a realization p is called ***regular*** if is satisfies
>
> $$\operatorname{rank}(\mathbf{R}(G,p)) = \max_{p \in \mathbb{R}^{dn}} \operatorname{rank}(\mathbf{R}(G,p))$$

While this definition might be confused with General position, they represent fundamentally different perspectives on a framework:

Difference between General and Regular

- **General position** focuses on the **geometry of the joints**; it requires that no $m+1$ points lie in an $(m-1)$-dimensional affine subspace, a condition that depends only on the point locations and is independent of the graph's edges, not depend on the structure of graph.
- **Regular realization** focuses on the **algebraic rank of the structure**; a realization p is regular if its rigidity matrix R_p attains the maximum possible rank k for that specific graph, meaning it inherently depends on the graph's combinatorial structure(connection).

Also, recall the independent framework.

Independent framework is Regular

For a graph G with $|V(G)| = n$ and $|E(G)| = m$, "The rows of $\mathbf{R}(G,p) \in \mathbb{R}^{m \times dn}$ are linearly independent" implies that it has **full row rank**, i.e.

$$\operatorname{rank}(\mathbf{R}(G,p)) = m$$

Then, $\mathbf{R}(G,p)$ has the maximum possible rank and hence it is **regular** as well.

Q. Is there any implication between them?

Prop 2.23) Regular implies General

> Given a framework (G, p), if p is regular, then p is general as well.

Proof

Prove the contrapositive: p is not general $\Rightarrow$ p is not regular.

Suppose that $p \in \mathbb{R}^{dn}$ is not general.

By the definition of general position, this means there exists a subset $S \subset V(G)$ of $d+1$ vertices that lie on an $(d-1)$-dimensional affine subspace H (and hence $H \cong \mathbb{R}^{d-1}$).

WLOG, let $S = \{p_1, \cdots, p_{d+1}\}$ where $p_i \in H$ for $i \in [d+1]$.
(Note: the other vertices $p_{d+2}, \cdots, p_n$ can be anywhere in $\mathbb{R}^d$).

Now, consider the submatrix M_S of $\mathbf{R}(K_n, p)$ corresponding to a subgraph of K_n induced by S(that is, K_{d+1}).

Recall theorem 2.16, that implies the maximum rank of rigidity matrix $\mathbf{R}(K_n, p)$ for d-dimensional framework is

$$\max_{q \in \mathbb{R}^{dn}} \operatorname{rank}(\mathbf{R}(K_n, q)) = dn - \binom{d+1}{2}.$$

So, since we only consider a subgraph K_{d+1}, the maximum possible rank of M_S at q is

$$\max_{q \in \mathbb{R}^{dn}} \operatorname{rank}(M_S(q)) = d(d+1) - \binom{d+1}{2} = \binom{d+1}{2} = \frac{d(d+1)}{2}.$$

and the rank for non-general realization,

$$\operatorname{rank}(M_S(p)) = (d-1)(d+1) - \binom{d-1+1}{2} = \frac{(d+2)(d-1)}{2}$$

Then, its difference be

$$\begin{aligned}\max_{q \in \mathbb{R}^{dn}} \operatorname{rank}(M_S(q)) - \operatorname{rank}(M_s(p)) &= \frac{d(d+1)}{2} - \frac{(d+2)(d-1)}{2} \\ &= \frac{(d^2+d) - (d^2+d-2)}{2} \\ &= 1 > 0\end{aligned}$$

Since the rank of the submatrix M_S at p is strictly less than its maximum possible rank, the rank of the entire rigidity matrix $\mathbf{R}(K_n, p)$ must also be strictly less than its maximum possible rank.

$$\operatorname{rank}(\mathbf{R}(K_n, p)) < \max_{q \in \mathbb{R}^{dn}} \operatorname{rank}(\mathbf{R}(K_n, q))$$

Therefore, by the definition of regular points, p is not regular. $\square$

Q. How about the converse? $\rightarrow$ NOT true.

Regular implies General: 3-Prism graph

Consider a 3-Prism graph G, that is

$$V(G) = \{a_1, a_2, a_3,\ b_1, b_2, b_3\}, \quad E(G) = \{a_1a_2, a_2a_3, a_3a_1,\ b_1b_2, b_2b_3, b_3b_1,\ a_1b_1, a_2b_2, a_3b_3\}$$

Let p be a realization on $\mathbb{R}^2$ defined by

$$\begin{aligned} p(a_1) = (1,0), \quad p(a_2) = (0,1), \quad p(a_3) = (-1,-1) \\ p(b_1) = (2,0), \quad p(b_2) = (0,2), \quad p(b_3) = (-2,-2) \end{aligned}$$

Then, since none of three vertices lie on a straight line and hence such a realization is general.

Next, its rigidity matrix is

$$\mathbf{R}(G,p) = \left[\begin{array}{cc|cc|cc|cc|cc|cc} 1 & -1 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & 2 & -1 & -2 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 2 & 1 & 0 & 0 & -2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 2 & -2 & -2 & 2 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 4 & -2 & -4 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 4 & 2 & 0 & 0 & -4 & -2 \\ \hline -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & -1 & -1 \end{array}\right] \in \mathbb{R}^{9\times(6\times 2)}$$

and its rank is 8.

Since

$$\max_{q\in\mathbb{R}^{2\times 6}} \operatorname{rank}(\mathbf{R}(G,q)) = 9 > 8 = \operatorname{rank}(\mathbf{R}(G,p)),$$

it is not regular.

---

3. Combinatorial aspects of rigidity

Motivation

Up to this point, our analysis has **focused on the rigidity of frameworks based on their specific geometric configurations**—that is, the exact coordinates of vertices in $\mathbb{R}^d$.

Now, a fundamental question arises: to what extent can these **properties be characterized solely by the underlying graph's connectivity**, independent of its specific realization?

So, we'll investigate the possibility of **treating rigidity as an intrinsic property** of the underlying graph, leading us to the study of generic frameworks where infinitesimal rigidity becomes purely combinatorial.

Introduce some definitions first.

Def 3.1) Indeterminate rigidity matrix

> Let K_n be the complete graph on n vertices. For each $i = 1, 2, \cdots, n$, we introduce a d-tuple of variables $p_i' = \left((p_i^{(1)})', \cdots, (p_i^{(d)})'\right)$ (d unkown variables).

Then, let

$$\mathbf{R}(n,d) = \begin{pmatrix} & & & & & \vdots & & & & & \\ 0 & \dots & 0 & p_i' - p_j' & 0 & \dots & 0 & p_j' - p_i' & 0 & \dots & 0 \\ & & & & & \vdots & & & & & \end{pmatrix}$$

be the matrix that is obtained from the rigidity matrix $\mathbf{R}(K_n, p)$ by replacing each $p_i^{(j)} \in \mathbb{R}$ with the unknown variable $(p_i^{(j)})'$. We call $\mathbf{R}(n,d)$ the d-dimensional ***indeterminate rigidity matrix of*** K_n.

Def 3.2) Generic framework

Let $V = \{v_1, \cdots, v_n\}$ and $p : V \to \mathbb{R}^d$ be a map.

We say that p is ***generic realization*** if the determinant of any submatrix of $\mathbf{R}(K_n, p)$ is zero only if the determinant of the corresponding submatrix of $\mathbf{R}(n,d)$ is (identically) zero.

A framework (G, p) is said to be ***generic*** if p is a generic map.

Algebraic independence & Geometric intuition

In other words, a realization p is said to be generic if elements of p are **algebraically independent.** That is, none of the polynomials $P(p_1, p_2, \cdots, p_n)$ satisfy

$$P(p_1, p_2, \cdots, p_n) = 0$$

over $\mathbb{Q}$.

Geometrically, **the point $p \in \mathbb{R}^{dn}$ does not lie on any non-trivial algebraic variety defined over $\mathbb{Q}$.**

Lem 3.3) Rank of generic frameworks

Let $p : V(G) \to \mathbb{R}^d$ be a generic map, Then, the rank of the rigidity matrix $\mathbf{R}(G,p)$ is equal to the maximum possible rank attainable by any realization of G in $\mathbb{R}^d$. That is,

$$\operatorname{rank}(\mathbf{R}(G,p)) = \max_{q \in \mathbb{R}^{dn}} \operatorname{rank}(R(G,q))$$

Proof

Let $k = \max_{q \in \mathbb{R}^{dn}} \operatorname{rank}(R(G,q))$ be the maximum rank.
By the definition of matrix rank, there exists at least one realization $q_0 \in \mathbb{R}^{dn}$ s.t. $\mathbf{R}(G, q_0)$ contains a $k \times k$ submatrix S with a non-zero determinant, i.e. $\det S(q_0) \neq 0$.

The determinant $\det S$ is a polynomial in the coordinates of the joints. Since there exists a point q_0 where $\det S(q_0) \neq 0$, this polynomial is not identically zero.

IAW definition 3.2, a map p is generic if the determinant of any submatrix is zero only if the corresponding determinant polynomial is identically zero.

Since our $k \times k$ determinant polynomial is not identically zero, it must be non-zero for any generic map p. Thus, $\operatorname{rank}(\mathbf{R}(G,p)) \geq k$.

As k is defined as the maximum possible rank, we can conclude that $\operatorname{rank}(\mathbf{R}(G,p) = k$. $\square$

Then, there is a result directly follows.

Cor) Genericity implies Regularity

> Every generic realization p of a graph G in $\mathbb{R}^d$ is a regular point.

Then, observe some examples of non-generic frameworks.

Example (Genericity of triangle frameworks)

We want to determine whether frameworks $(G,p),(G,q)$ in $\mathbb{R}^2$ is generic where

$$V(G) = \{v_1, v_2, v_3\}$$

with distinct realizations p, q where

$$p_1 = (0,0),\ p_2 = (1,0),\ p_3 = (2,0) \quad \text{and} \quad q_1 = (0,0),\ q_2 = (1,0),\ q_3 = (0,1)$$

Before algebraically determining them, we can guess.

1. p seems NOT generic because three points are colinear.
2. q seems generic because three points are not colinear.

Let's check.

First, construct an indeterminate rigidity matrix with $n = 3$.

$$\mathbf{R}(3,2) = \left(\begin{array}{cc|cc|cc} x_1 - x_2 & y_1 - y_2 & x_2 - x_1 & y_2 - y_1 & 0 & 0 \\ \hline x_1 - x_3 & y_1 - y_3 & 0 & 0 & x_3 - x_1 & y_3 - y_1 \\ \hline 0 & 0 & x_2 - x_3 & y_2 - y_3 & x_3 - x_2 & y_3 - y_2 \end{array}\right)$$

and construct $\mathbf{R}(K_3,p)$.

$$\mathbf{R}(K_3,p) = \left(\begin{array}{cc|cc|cc} 0-1 & 0-0 & 1-0 & 0-0 & 0 & 0 \\ \hline 0-2 & 0-0 & 0 & 0 & 2-0 & 0-0 \\ \hline 0 & 0 & 1-2 & 0-0 & 2-1 & 0-0 \end{array}\right) = \left(\begin{array}{cc|cc|cc} -1 & 0 & 1 & 0 & 0 & 0 \\ \hline -2 & 0 & 0 & 0 & 2 & 0 \\ \hline 0 & 0 & -1 & 0 & 1 & 0 \end{array}\right)$$

Taking submatrix $\mathbf{R}_{(145)}$of $\mathbf{R}(K_3,p)$ by taking column 1, 4 and 5. Then,

$$\det \mathbf{R}_{(145)} = \det \begin{pmatrix} -1 & 0 & 0 \\ -2 & 0 & 2 \\ 0 & 0 & 1 \end{pmatrix} = 0$$

However, a determinant of corresponding submatrix $\mathbf{R}(3,2)_{145}$ is

$$\begin{aligned} \det \mathbf{R}(3,2)_{(145)} &= \det \begin{pmatrix} x_1 - x_2 & y_2 - y_1 & 0 \\ x_1 - x_3 & 0 & x_3 - x_1 \\ 0 & y_2 - y_3 & x_3 - x_2 \end{pmatrix} \\ &= (x_1 - x_3)[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)] \end{aligned}$$

This is not necessarily zero (may not be zero by choosing coordinates properly) and hence, p **is NOT a generic realization.** ($\rightarrow$ We correctly guessed)

Geometrically, as we've guessed, p lies on a 6-dimensional algebraic variety, a straight line**, and hence non-generic.

Now, again, to determine whether q is generic or not, construct $\mathbf{R}(K_3, q)$.

$$\mathbf{R}(K_3, q) = \left(\begin{array}{cc|cc|cc} 0-1 & 0-0 & 1-0 & 0-0 & 0 & 0 \\ \hline 0-0 & 0-1 & 0 & 0 & 0-0 & 1-0 \\ \hline 0 & 0 & 1-0 & 0-1 & 0-1 & 1-0 \end{array}\right) = \left(\begin{array}{cc|cc|cc} -1 & 0 & 1 & 0 & 0 & 0 \\ \hline 0 & -1 & 0 & 0 & 0 & 1 \\ \hline 0 & 0 & 1 & -1 & -1 & 1 \end{array}\right)$$

Taking submatrix $\mathbf{R}_{(145)}$of $\mathbf{R}(K_3, q)$ by taking column 1, 4 and 5. Then,

$$\det \mathbf{R}_{(145)} = \det \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & -1 \end{pmatrix} = 0$$

However, a determinant of corresponding submatrix $\mathbf{R}(3,2)_{145}$ is

$$\begin{aligned} \det \mathbf{R}(3,2)_{(145)} &= \det \begin{pmatrix} x_1 - x_2 & y_2 - y_1 & 0 \\ x_1 - x_3 & 0 & x_3 - x_1 \\ 0 & y_2 - y_3 & x_3 - x_2 \end{pmatrix} \\ &= (x_1 - x_3)[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)] \end{aligned}$$

This is not necessarily zero (may not be zero by choosing coordinates properly) and hence, q **is NOT a generic realization.** ($\rightarrow$ Our guess was Wrong!)

Geometrically, **each points q lies on a 6-dimensional algebraic variety** $x_1 - x_3 = 0$ and hence non-generic.

In summary,

Relation between position

$$\text{Generic} \subset \text{Regular} \subset \text{General} \subset \text{Normal}$$

Next, there are some fundamental facts regarding this definition of generic.

Thm 3.3) Set theoretic properties of non-generic realizations

> The set of all non-generic realizations $p : V \rightarrow \mathbb{R}^d$ where $V = \{v_1, \cdots, v_n\}$ is a **closed set of measure zero.**

Proof

Let $p : V(K_n) \rightarrow \mathbb{R}^d$ be a realization of K_n and it can also be considered as a vector $p \in \mathbb{R}^{dn}$ (Since the genericity is only defined for realizations, we can assume a framework with complete graph K_n).

Let $\mathcal{S}$ be the set of all square submatrix of $\mathbf{R}(K_n, p)$.

Also, given $M \in \mathcal{S}$, define a map $f_M : \mathbb{R}^{dn} \to \mathbb{R}\ ;\ \ p \mapsto \det M(p)$. Here, since each determinant is the sums and products of elements of matrix and each elements of rigidity matrix is polynomial, f_M is a polynomial for all $M \in \mathcal{S}$.

Suppose that p is non-generic. Then, by the definition of generic realization, $\exists M \in \mathcal{S}$ s.t. $f_M(p) = 0$ but f_M is not identically zero (i.e. $\exists q \in \mathbb{R}^{dn}$ s.t. $f_M(q) \neq 0$).

So, the set of non-generic realizations can be described by

$$\mathcal{G} = \bigcup_{\substack{M \in \mathcal{S} \\ f_M \not\equiv 0}} \{p \in \mathbb{R}^{dn} : f_M(p) = 0\}.$$

Now, claim that $\mathcal{G}$ is a closed set.

Note that $\{p \in \mathbb{R}^{dn} : f_M(p) = 0\} = f_M^{-1}(\{0\})$. Since f_M is continuous and a one-point set $\{0\}$ is closed, its inverse image $f_M^{-1}(\{0\})$ is closed as well.

Also, since the number of submatrices of $\mathbf{R}(K_n, p)$ is finite, $\mathcal{G}$ is a finite union of closed set and hence closed.

Next, claim that $\mathcal{G}$ is measure zero.

Note a lemma first: For a given nontrivial polynomial $f : \mathbb{R}^N \to \mathbb{R}$, its zero set $f^{-1}(\{0\}) = \{x \in \mathbb{R}^N : f(x) = 0\}$ has Lebesgue measure zero.

Then, by the subadditivity of measure,

$$\mu(\mathcal{G}) = \mu\left(\bigcup_M f_M^{-1}(\{0\})\right) \leq \sum_M \mu(f_M^{-1}(\{0\})) = 0$$

and hence $\mu(\mathcal{G}) = 0$ as well. $\square$

Thm 3.4) Combinatorial Rigidity Theorem

> For a fixed dimension d and a graph G with $n(G) \geq d$, TFAE:
>
> 1. (G, p) is infinitesimally rigid (or independent, or isostatic) for at least one injective map $p : V(G) \to \mathbb{R}^d$.
> 2. (G, p) is infinitesimally rigid (or independent, or isostatic) for every d-dimensional generic realization p.

Intuitively, success in a single normal configuration implies success in almost every possible configuration.

The three structural properties (**infinitesimal rigidity**, **independence**, and **isostaticity**) only depend on the **underlying graph** G; specifically its **combinatorial structure**.

Proof

$(1) \Rightarrow (2)$: Suppose that (G, p) is infinitesimally rigid for an injective $p : V(G) \to \mathbb{R}^d$ (not necessarily generic).

By theorem 2.16, for infinitesimally rigid framework,

$$\operatorname{rank}(\mathbf{R}(G,p)) = dn - \binom{d+1}{2}$$

where $n(G) = n$.

Let $\hat{p}$ be a d-dimensional generic framework for a graph G. By lemma 3.3, the rigidity matrix of $\hat{p}$ has the maximum rank and hence $\operatorname{rank}(\mathbf{R}(G,\hat{p})) \geq \operatorname{rank}(\mathbf{R}(G,p))$.

Again, by theorem 2.16, a rank of rigidity matrix of d-dimensional framework cannot exceed $dn - \binom{d+1}{2}$ and hence $\operatorname{rank}(\mathbf{R}(G,\hat{p})) \leq dn - \binom{d+1}{2}$.

Combining the equation and inequalities, we obtain

$$dn - \binom{d+1}{2} = \operatorname{rank}(\mathbf{R}(G,p)) \leq \operatorname{rank}(\mathbf{R}(G,\hat{p})) \leq dn - \binom{d+1}{2}$$

Thus, $\operatorname{rank}(\mathbf{R}(G,\hat{p})) = dn - \binom{d+1}{2}$ and hence we can conclude that $(G,\hat{p})$ is infinitesimally rigid by theorem 2.16.

$(2) \Rightarrow (1)$: Suppose that (G,p) is infinitesimally rigid for every generic realization p.

Existence of p: Since the set of generic realizations is open dense set in $\mathbb{R}^{dn}$, there exists a map $p : V(G) \to \mathbb{R}^d$ s.t. (G,p) is infinitesimally rigid.

Injectivity: Also, since generic framework is general, the vertices never overlap and hence p must be injective.

Thus, for any generic realization p, it is an injective realization that satisfies (G,p) is infinitesimally rigid.

This proof can be applied directly for independence or isostaticity instead of infinitesimal rigidity. $\square$

Thm 3.5) Equivalence of rigidity

> Infinitesimal rigidity is equivalent to rigidity for generic frameworks.

Proof

Since we've already proved that infinitesimal rigidity implies rigidity(even if the framework is non-generic), we only need to converse.

Claim 1: $\operatorname{rank}(\mathbf{R}(G,\cdot))$ is lower semicontinuous.

Note that $\operatorname{rank}(\mathbf{R}(G,p)) = k$.

By the definition of matrix rank, there exists at least one $k \times k$ submatrix with nonzero determinant denoted by $M(p)$.

Since the entries of $\mathbf{R}(G,p)$ consists of linear differences of the vertex coordinates, they are polynomials and hence continuous. Also, since a determinant is just sums and products of the entries, a map $h : q \mapsto \det(M(q))$ is continuous mapping from $\mathbb{R}^{dn} \to \mathbb{R}$.

Since $\mathbb{R} \setminus \{0\}$ is open, its inverse image under h, $h^{-1}(\mathbb{R} \setminus \{0\})$ is an open set in $\mathbb{R}^{dn}$.

Thus, there exists a neighborhood U of p s.t. for all configurations $q \in U$, $\det(M(q)) \neq 0$.

Because this $k \times k$ submatrix maintains a nonzero determinant throughout U, the overall matrix $\mathbf{R}(G, q)$ must have a rank of at least k for all $q \in U$.

Thus, $\operatorname{rank}(\mathbf{R}(G, q)) \geq \operatorname{rank}(\mathbf{R}(G, p))$ and hence it is lower semicontinuous.

Claim 2: Complete the proof

Next, let p be a regular point and let k be the maximum possible rank of the rigidity matrix. Then, by the definition of regular point,

$$\operatorname{rank}(\mathbf{R}(G, p)) = k := \max_{q \in \mathbb{R}^{dn}} \operatorname{rank}(\mathbf{R}(G, q))$$

Since $\operatorname{rank}(\mathbf{R}(G, p)) = k$ is the maximum, we have

$$\operatorname{rank}(\mathbf{R}(G, q)) \leq k = \operatorname{rank}(\mathbf{R}(G, p)), \quad \forall q \in U$$

So, together with the lower semicontinuity,

$$\operatorname{rank}(\mathbf{R}(G, q)) = k, \quad \forall q \in U. \qquad \text{(constant rank)}$$

Then, by the constant rank theorem, the configuration space near p, i.e.

$$\mathcal{C}(G, p) \cap U = f_G^{-1}(f_G(p)) \cap U$$

be a smooth regular submanifold of $\mathbb{R}^{dn}$ and the dimension is

$$\dim(f_G^{-1}(f_G(p)) \cap U) = dn - k$$

Also, note that $f_K^{-1}(f_K(p))$, a set of all realizations congruent to p is an orbit of p under an isometry group $E(d)$ in the view of group action, i.e.

$$f_K^{-1}(f_K(p)) \cap U = \mathcal{O}_p \cap U$$

Assuming the framework is general, the dimension of such an orbit is exactly $\binom{d+1}{2}$.

Then, if (G, p) is rigid, then

$$f_G^{-1}(f_G(p)) \cap U = f_K^{-1}(f_K(p)) \cap U$$

and hence their dimension must be equal.

$$dn - k = \binom{d+1}{2}$$

By rearranging this, we have $k = \operatorname{rank}(\mathbf{R}(G, p)) = dn - \binom{d+1}{2}$ and hence, (G, p) is infinitesimally rigid by theorem 2.17.

Therefore, as the equivalence holds for regular points, it automatically holds for generic frameworks as well. $\square$

Def 3.6) Generically rigid/independent/isostatic

> A graph G is called ***generically rigid/independent/isostatic in dimension*** d if d-dimensional generic realizations of G are infinitesimally rigid/independent/isostatic.

Here, by theorem 3.5, A graph G is generically rigid if a generic realizations of G is rigid ($\because$ inifintiesimally rigid $\equiv$ rigid for generic).

Note: Generically d-isostatic graphs can be constructed using a sequence of Henneberg construction, which are inductive construction techniques.

First, introduce the definitions of Henneberg construction, which consists of two extension techniques.

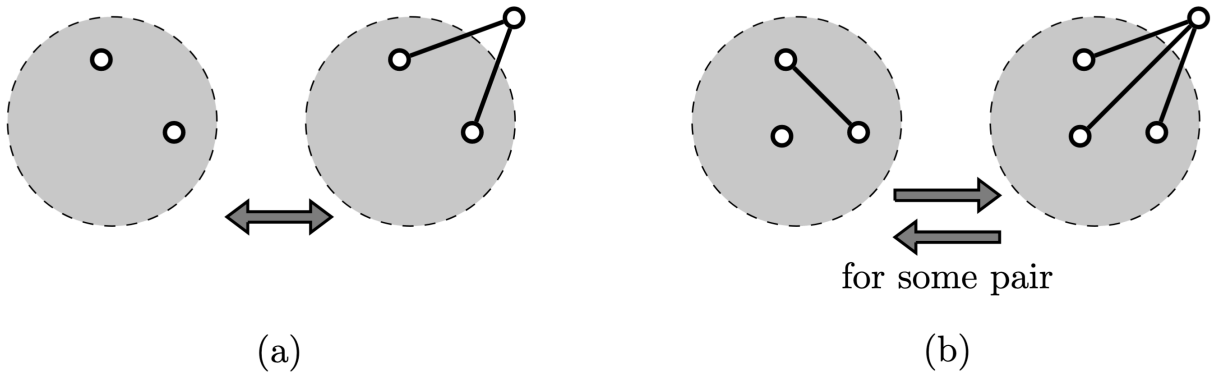
Def 3.7) Vertex addition (0-extension)

> Let G be a graph, $U \subseteq V(G)$ with $|U| = d$ and $v \notin V(G)$. Then the graph $\widehat{G}$ with $V(\widehat{G}) = V(G) \cup \{v\}$ and $E(\widehat{G}) = E(G) \cup \{\{v, u\} : u \in U\}$ is called ***a vertex d-addition*** (by v) of G.

Intuitively, a vertex d-addition is the process of expanding a graph by introducing **one new vertex** and connecting it to exactly d **distinct existing vertices** via d new edges.

Def 3.8) Edge split (1-extension)

> Let G be a graph, and let $U \subseteq V(G)$ with $|U| = d + 1$ and $\{u_1, u_2\} \in E(G)$ for some $u_1, u_2 \in U$. Further, let $v \notin V(G)$. Then, the graph $\hat{G}$ with $V(\hat{G}) = V(G) \cup \{v\}$ and $E(\hat{G}) = (E(G) \setminus \{u_1u_2\}) \cup \{vu : u \in U\}$ is called an ***edge d-split (on $u_1, u_2; v$)*** of G.



This figure illustrates (a) the vertex 2-addition and (b) the edge 2-split.

Henneberg construction serves as a fundamental building block for inductive constructions because it preserves the structural properties such as generic rigidity, as stated in the following theorems.

Thm 3.9) Vertex Addition Theorem

> A vertex d-addition of a generically d-isostatic graph is generically d-isostatic. Conversely, deleting a vertex of degree d from a generically d-isostatic graph results in a generically d-isostatic graph.

Thm 3.10) Edge Split Theorem

> An edge d-split of a generically d-isostatic graph is generically d-isostatic. Conversely, if one deletes a vertex v of valence $d + 1$ from a generically d-isostatic graph, then one may add an edge between one of the pairs of vertices adjacent to v so that the resulting graph is generically d-isostatic.

Henneberg gave the following characterization of generically 2-isostatic graphs.

Thm 3.11) Henneberg's Construction Theorem

> A graph is generically 2-isostatic if and only if it may be constructed from a single edge by a sequence of vertex 2-additions and edge 2-splits.

Using Henneberg's construction theorem, we can prove the following characterization of generically 2-isostatic graphs due to Laman.

Thm 3.12) Laman's Theorem

> A graph G with $|V(G)| \geq 2$ is generically 2-isostatic if and only if
>
> 1. $|E(G)| = 2|V(G)| - 3$
> 2. $|E(H)| \geq 2|V(H)| - 3$ for all $H \subseteq G$ with $|V(H)| \geq 2$

Proof

Direction 1: 2-tree ⇒ Laman's Conditions

We show by induction on the construction sequence that every 2-tree G satisfies the Laman conditions.

Recall: A *2-tree* is a graph that can be built from a single edge K_2 by a sequence of:

- **Vertex 2-additions:** Add a new vertex v connected to exactly 2 existing vertices.
- **Edge 2-splits:** Remove an edge $\{u_1, u_2\}$, add a new vertex v connected to u_1, u_2, and one additional vertex u_3.

i) Base case: K_2 has $|V(K_2)| = 2$ and $|E(K_2)| = 1 = 2(2) - 3$. The only subgraph with ≥ 2 vertices is K_2 itself, so condition (2) holds trivially. ✓

ii) Inductive step: Suppose G' is a 2-tree satisfying Laman's conditions.

(a) Vertex 2-addition. Add vertex v with edges to $u_1, u_2 \in V(G')$, forming G.

Then

$$|V(G)| = |V(G')| + 1 \quad \text{and} \quad |E(G)| = |E(G')| + 2 = (2|V(G')| - 3) + 2 = 2|V(G)| - 3$$

and hence condition (1) holds.

For condition (2), take any $H \subseteq G$ with $|V(H)| \geq 2$:

First, if $v \notin V(H)$, then $H \subseteq G'$, so $|E(H)| \leq 2|V(H)| - 3$ by the inductive hypothesis.

Secondly, if $v \in V(H)$, then let $H' = H - v$.

Then $|E(H)| \leq |E(H')| + 2$ (since v has degree 2 in G) and $|V(H)| = |V(H')| + 1$.

Here, if $|V(H')| \geq 2$, then $|E(H)| \leq |E(H')| + 2 \leq (2|V(H')| - 3) + 2 = 2|V(H)| - 3$.

Else if $|V(H')| = 1$, then $|E(H)| \leq 1 \leq 2(2) - 3 = 1$. ✓

(b) Edge 2-split on $\{u_1, u_2\}$ via v and u_3. Remove $\{u_1, u_2\}$ from G', add v with edges to u_1, u_2, u_3, forming G.

Then

$$|V(G)| = |V(G')| + 1 \quad \text{and} \quad |E(G)| = |E(G')| - 1 + 3 = (2|V(G')| - 3) + 2 = 2|V(G)| - 3$$

and hence condition (1) holds.

For condition (2), take any $H \subseteq G$ with $|V(H)| \geq 2$:

- If $v \notin V(H)$: the edges of H are a subset of $E(G') \setminus \{\{u_1, u_2\}\} \subseteq E(G')$. So H is a subgraph of G', and by the inductive hypothesis, $|E(H)| \leq 2|V(H)| - 3$. ✓
- If $v \in V(H)$: let $H' = H - v$. The edges of H consist of edges of H' (which are edges of G' within $V(H')$, minus possibly $\{u_1, u_2\}$) plus edges from v (at most 3). Let e denote the number of edges of G' within $V(H')$, so $e \leq 2|V(H')| - 3$ by the inductive hypothesis when $|V(H')| \geq 2$.
  - If $u_1, u_2 \in V(H')$ (so $\{u_1, u_2\} \in E(G')$ is an edge within $V(H')$): the edge $\{u_1, u_2\}$ was removed in the split, so H' has at most $e - 1$ edges, and v has at most 3 edges to $V(H')$. Thus $|E(H)| \leq (e - 1) + 3 \leq (2|V(H')| - 3) - 1 + 3 = 2|V(H)| - 3$. ✓
  - If not both $u_1, u_2 \in V(H')$: then at most 2 of $\{u_1, u_2, u_3\}$ lie in $V(H')$, so v has at most 2 edges to $V(H')$. Thus $|E(H)| \leq e + 2 \leq (2|V(H')| - 3) + 2 = 2|V(H)| - 3$. ✓
  - If $|V(H')| = 1$: $|E(H)| \leq 1 \leq 2(2) - 3$. ✓

Thus every 2-tree satisfies Laman's conditions. ■

Direction 2: Laman's Conditions ⇒ 2-tree

We proceed by **induction** on $|V(G)|$.

i) Base case: $|V(G)| \leq 3$ is easily checked.

ii) Inductive step: Assume $n \geq 3$ and that all graphs on n or fewer vertices satisfying Laman's conditions are 2-trees. Let G with $|V(G)| = n + 1$ satisfy the Laman conditions. As argued previously, G contains a vertex of valence 2 or 3.

Case 1: a_0 has valence 2

Let $\{a_0, a_1\}, \{a_0, a_2\} \in E(G)$. Let $G' = G - a_0$ (remove a_0 and its two incident edges).

Laman condition (2) for G': For any $U \subseteq V(G')$ with $|U| \geq 2$, the subgraph of G' induced by U is also a subgraph of G, so:

$$|E(G'[U])| = |E(G[U])| \leq 2|U| - 3.$$

Laman condition (1): $|E(G')| = |E(G)| - 2 = 2|V(G)| - 5 = 2|V(G')| - 3$. ✓

By the induction hypothesis, G' is a 2-tree. Since G is a vertex 2-addition of G', it is also a 2-tree. ■

Case 2: a_0 has valence 3

Let $\{a_0, a_1\}, \{a_0, a_2\}, \{a_0, a_3\} \in E(G)$. Let $G' = G - a_0$ (remove a_0 and its three incident edges).

Goal: Show that one of $\{a_1, a_2\}$, $\{a_1, a_3\}$, or $\{a_2, a_3\}$ can be added to G' without violating Laman's conditions. Then:

- G' plus that edge satisfies Laman, so by induction it is a 2-tree.
- G is an edge 2-split of this graph, so G is also a 2-tree.

Key claim: For one of the three pairs $\{a_i, a_j\}$, we have $|E(G'[U])| < 2|U| - 3$ for all $U \subset V(G')$ containing that pair.

First, handle the case where all three are in U.

Suppose there exists $U \subset V(G')$ with $a_1, a_2, a_3 \in U$ and $|E(G'[U])| = 2|U| - 3$. Then:

$$|E(G[U \cup \{a_0\}])| = |E(G'[U])| + 3 = (2|U| - 3) + 3 = 2|U| = 2|U \cup \{a_0\}| - 2,$$

which violates Laman condition (2) for G. So for any $U \subset V(G')$ containing a_1, a_2, a_3:

$$|E(G'[U])| < 2|U| - 3.$$

Now suppose the claim fails for all three pairs. Then there exist tight sets:

- $U_1 \subset V(G') - \{a_1\}$ containing a_2, a_3 with $|E(G'[U_1])| = 2|U_1| - 3$
- $U_2 \subset V(G') - \{a_2\}$ containing a_1, a_3 with $|E(G'[U_2])| = 2|U_2| - 3$
- $U_3 \subset V(G') - \{a_3\}$ containing a_1, a_2 with $|E(G'[U_3])| = 2|U_3| - 3$

(Each U_i cannot contain all three of a_1, a_2, a_3 by the argument above, so $a_i \notin U_i$.)

Step 1: Show $|U_i \cap U_j| = 1$ for all distinct i, j.

Let $U = U_1 \cup U_2$. By the **submodular inequality**:

$$|E(G'[U])| \geq |E(G'[U_1])| + |E(G'[U_2])| - |E(G'[U_1 \cap U_2])|.$$

(This holds because $|E(G'[A])| + |E(G'[B])|$ counts edges in $G'[A \cap B]$ twice, and misses edges between $A \setminus B$ and $B \setminus A$.)

Assuming $|U_1 \cap U_2| > 1$, we'd have $|E(G'[U_1 \cap U_2])| \leq 2|U_1 \cap U_2| - 3$, so:

$$|E(G'[U])| \geq (2|U_1| - 3) + (2|U_2| - 3) - (2|U_1 \cap U_2| - 3) = 2|U| - 3.$$

But U contains a_1, a_2, a_3, forcing $|E(G'[U])| < 2|U| - 3$. **Contradiction.**

Hence $|U_1 \cap U_2| = 1$, and by symmetry, $|U_i \cap U_j| = 1$ for all $\{i, j, k\} = \{1, 2, 3\}$.

Specifically, $U_i \cap U_j = \{a_k\}$ (the one vertex they share).

Step 2: Derive a final contradiction.

Let $U = U_1 \cup U_2 \cup U_3$. Since the pairwise intersections are singletons:

$$|U| = |U_1| + |U_2| + |U_3| - 3.$$

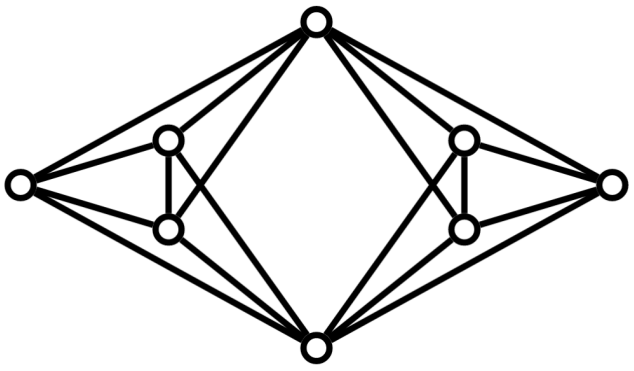
Since $U_i \cap U_j = \{a_k\}$ is a single vertex, no edge can belong to both $E(G'[U_i])$ and $E(G'[U_j])$ (such an edge would need both endpoints in $U_i \cap U_j$, but a single vertex has no edges). Therefore the edge sets $E(G'[U_1]), E(G'[U_2]), E(G'[U_3])$ are pairwise disjoint, and:

$$\begin{aligned} |E(G'[U])| &\geq |E(G'[U_1])| + |E(G'[U_2])| + |E(G'[U_3])| \\ &= (2|U_1| - 3) + (2|U_2| - 3) + (2|U_3| - 3) = 2|U| - 3. \end{aligned}$$

But U contains a_1, a_2, a_3, so $|E(G'[U])| < 2|U| - 3$. **Contradiction.** ■

Therefore, the assumption that all three pairs fail is false. One of the pairs $\{a_i, a_j\}$ can be added to G' while preserving Laman's conditions, completing the induction. □

As we mentioned in History of Rigidity theory, generically d-isostatic graphs cannot be characterized combinatorially by directly extending Laman's theorem because of the following counterexample (Double Banana graph).



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4. Rigidity of Polytopes

In History of Rigidity theory, we've mentioned that the Cauchy's rigidity theorem is one of the first celebrated results in the rigidity theory.

Our next goal is proving the Alexandrov's theorem, which is slight generalized version of the Cauchy's rigidity theorem.

As the proof of such a theorem utilizes techniques from static rigidity, let's shift the mathematical perspective from motions/infinitesimal motions to the concept of static rigidity.

4.1. Basic of Static rigidity

Intuitively, static rigidity can be described as follows:

- Instead of asking "Can this framework bend?", static rigidity asks:
  "If I apply an external load by pushing and pulling on the joints, can the bars internally absorb those forces without the structure collapsing?".
- The bars absorb these forces by either pulling inward (tension) or pushing outward (compression).

You can't apply just any force to test rigidity. If you push the whole framework in one direction, it will just slide through space. We only care about **equilibrium loads**, which must satisfy two physical rules:

- **No Translation:** All the applied force vectors must completely cancel each other out so the structure doesn't drift.
- **No Rotation:** The applied forces must not create a net twisting or rotational motion on the framework.

Formally, the equilibrium load is defined by:

Def 4.1) Load, Equilibrium load

> Let (G,p) be a framework in $\mathbb{R}^d$ with $V(G)=\{v_1,\cdots,v_n\}$. A ***load*** on (G,p) is a function $l:V(G)\to\mathbb{R}^d$, where for $i=1,2,\cdots,n$, the vector $l(v_i)$ represents a force applied to the joint (v_i,p_i) of (G,p).
>
> A load l on (G,p) is called an ***equilibrium load*** if l satisfies
>
> 1. $\sum_{i=1}^n l_i=0$
> 2. $\sum_{i=1}^n((l_i)_j(p_i)_k-(l_i)_k(p_i)_j)=0$ for all $1\le j<k\le d$,
> where l_i denotes the vector $l(v_i)$ for all i.

Next, if an equilibrium load is applied, the framework must "resolve" it.

- A resolution assigns a specific tension (a negative value) or compression (a positive value) to every single bar in the framework.
- At every individual joint, the external force applied to it (li) plus the internal pushes and pulls from the connected bars must perfectly balance out to zero.
- If a framework can successfully resolve *every* possible equilibrium load, it is definitively **statically rigid**.

This is formally stated by the following definition:

Def 4.2) Resolution

> Let l be an equilibrium load on a framework (G,p) in $\mathbb{R}^d$ with $V(G)=\{v_1,\cdots,v_n\}$. A ***resolution*** of l by (G,p) is a function $\omega:E(G)\to\mathbb{R}$ s.t. at each joint (v_i,p_i) of (G,p), we have
>
> $$\sum_{j \text{ with } v_iv_j\in E(G)}\omega_{ij}(p_i-p_j)+l_i=0 \quad\cdots\quad (\dagger)$$
>
> where ω_{ij} denotes $\omega(v_iv_j)$ for all $v_iv_j\in E(G)$.

Here, the scalars ω_{ij} represent tensions $(\omega_{ij}<0)$ and compressions $(\omega_{ij}>0)$ in the bars v_iv_j in of (G,p), so that the bar forces reach an equilibrium with l_i at each joint (v_i,p_i).

Note that if we identify l and ω with a column vector in $\mathbb{R}^{dn}$ and $\mathbb{R}^{|E(G)|}$, respectively, then the equation $(\dagger)$ can be written in a compact form in terms of rigidity matrix.

$$\mathbf{R}(G,p)^\top\omega=l$$

Fundamental Equilibrium load

Given two joints $(v_h, p_h), (v_k, p_k)$ of (G, p), the fundamental equilibrium load is of the form

$$(F_{hk})^\mathsf{T} = (0, \cdots, 0, p_h - p_k, 0, \cdots, 0, p_k - p_h, 0, \cdots, 0)$$

Proof) While the first condition(no translational move) of the equilibrium load is trivial, the second condition(no rotational move) is less trivial.

Let $l = F_{hk}$. Then, $l_h = p_h - p_k$ and $l_k = p_k - p_h$.

So,

$$\begin{aligned}\sum_{i=1}^{n}[(l_i)_\alpha(p_i)_\beta - (l_i)_\beta(p_i)_k] &= [(l_h)_\alpha(p_h)_\beta - (l_h)_\beta(p_h)_\alpha] + [(l_k)_\alpha(p_k)_\beta - (l_k)_\beta(p_k)_\alpha] \\ &= [(p_h - p_k)_\alpha(p_h)_\beta - (p_h - p_k)_\beta(p_h)_\alpha] + [(p_k - p_h)_\alpha(p_h)_\beta - (p_k - p_h)_\beta(p_h)_\alpha] \\ &= 0 \quad (\text{Cancelled out})\end{aligned}$$

Hence, proved. □

Further, if $v_iv_j \in E(G)$, then $(F_{ij})^\mathsf{T}$ is the row vector of $\mathbf{R}(G, p)$ corresponds to $v_iv_j \in E(G)$ and F_{ij} is clearly resolved by the bar $\{(v_i, p_i), (v_j, p_j)\}$.

Def 4.3) Statical Rigidity

> A framework (G, p) is statically rigid if every equilibrium load on (G, p) has a resolution by (G, p).

Note: If (G, p) is statically rigid, then the fundamental equilibrium load F_{hk} has a resolution by (G, p) for every pair (h, k) of joints of (G, p) even if $v_hv_k \notin E(G)$.

Q. Is the converse True? $\rightarrow$ Not in general, but holds for normal frameworks.

Thm 4.4) Equivalence with Infinitesimal rigidity

> Assume that p is a normal position, i.e. $p_1, \cdots, p_n$ span $\mathbb{R}^d$. Then, a framework (G, p) is infinitesimally rigid if and only if (G, p) is statically rigid.

Proof

Claim: The vectors F_{hk}, $1 \le h < k \le n$ generate the entire space of equilibrium loads on (G, p).

We'll prove this claim by dimension comparison.

Note that the vectors $(F_{hk})^\mathsf{T}$ for every possible pair (h, k) s.t. $1 \le h < k \le n$ exactly correspond to the rows of $\mathbf{R}(K_n, p)$.

Since (K_n, p) is always infinitesimally rigid, we have $\text{rank}\mathbf{R}(K_n, p) = dn - \binom{d+1}{2}$ by theorem 2.17. This implies that the space generated by the rows of $\mathbf{R}(K_n, p)$, i.e. the space generated by $(F_{hk})^\mathsf{T}$ has dimension $dn - \binom{d+1}{2}$.

Next, recall that a load l is an equilibrium load if

1. $\sum_{i=1}^{n} l_i = 0$
2. $\sum_{i=1}^{n}((l_i)_j(p_i)_k - (l_i)_k(p_i)_j) = 0$ for all $1 \le j < k \le d$.

Here, since $l_i \in \mathbb{R}^d$ for all i, the first equation is d linear equations with n variables, i.e.

$$l_1 + \cdots + l_n = \begin{bmatrix} l_{11} \\ l_{12} \\ \vdots \\ l_{1d} \end{bmatrix} + \begin{bmatrix} l_{21} \\ l_{22} \\ \vdots \\ l_{2d} \end{bmatrix} + \cdots + \begin{bmatrix} l_{n1} \\ l_{n2} \\ \vdots \\ l_{nd} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\Longrightarrow \begin{cases} l_{11} + l_{21} + \cdots + l_{n1} = 0 \\ l_{12} + l_{22} + \cdots + l_{n2} = 0 \\ \vdots \\ l_{1d} + l_{2d} + \cdots + l_{nd} = 0 \end{cases}$$

Here, for each equation, while $n-1$ variables can be selected freely, the last variable is forced and hence d variables are forced in total.

So, the frameworks loses d DOFs for translation.

Similarly, for rotation, each equation is determined w.r.t. planes and exactly one variable is forced for each plane. So, the number of forced variables equals the number of independent planes.

Since each plane is determined by two vectors, there are $\binom{d}{2}$ independent planes and hence the framework loses $\binom{d}{2}$ DOFs for rotation.

Thus, the space of every equilibrium load has dimension

$$dn - d - \binom{d}{2} = dn - \binom{d+1}{2}.$$

Thus, we can find that the dimensions of the space generated by F_{hk}s and the space of equilibrium loads are equal.

Since each F_{hk} is an equilibrium load, we can conclude that the space generated by F_{hk}s is the space of equilibrium loads.

$(\Leftarrow)$ Suppose that (G,p) is statically rigid.

By claim, we can find that there are $dn - \binom{d+1}{2}$ independent rows of $\mathbf{R}(G,p)$ if (G,p) is statically rigid. So, $\mathrm{rank}\mathbf{R}(G,p)^\mathsf{T}$(row rank) is exactly $dn - \binom{d+1}{2}$.

Since $\mathrm{rank}\mathbf{R}(G,p)^\mathsf{T} = \mathrm{rank}\mathbf{R}(G,p)$, we can conclude that

$$\mathrm{rank}\mathbf{R}(G,p) = dn - \binom{d+1}{2}$$

and hence (G,p) is infinitesimally rigid.

$(\Rightarrow)$ Suppose that (G,p) is infinitesimally rigid, i.e.

$$\mathrm{rank}\mathbf{R}(G,p) = dn - \binom{d+1}{2}$$

Since $\mathrm{rank}\mathbf{R}(G,p) = \mathrm{rank}\mathbf{R}(G,p)^\mathsf{T}$, we have $\mathrm{rank}\mathbf{R}(G,p)^\mathsf{T} = dn - \binom{d+1}{2}$.

Note that the rows of $\mathbf{R}(G,p)$ generate the space of equilibrium loads of dimension $dn - \binom{d+1}{2}$.

This implies that the row space of $\mathbf{R}(G,p)$ is exactly the entire space of equilibrium loads, i.e. given an equilibrium load l, it can be expressed as the linear combination of the rows of $\mathbf{R}(G,p)$.

Formally stating, for any given equilibrium load l, we can always find a resolution ω s.t.

$$\mathbf{R}(G,p)^{\mathsf{T}}\omega = l.$$

By definition, (G,p) is statically rigid. $\square$

Def 4.5) Stress, Self-stress

> Given a framework (G,p), a function $\omega : E(G) \to \mathbb{R}$ is called a ***stress*** of (G,p). Equivalently, a stress of (G,p) is a resolution of a load on (G,p).
>
> A resolution of the zero-load (i.e. no external load $l = 0$) is called a ***self-stress*** (=static stress) of (G,p).
>
> So, if we identify a stress of (G,p) with a vector in $\mathbb{R}^{|E(G)|}$, then a vector $\omega \in \mathbb{R}^{|E(G)|}$ is a self-stress of (G,p) if $\mathbf{R}(G,p)^{\mathsf{T}}\omega = 0$.

In other words, the set of all self-stresses is $\ker(\mathbf{R}(G,p)^{\mathsf{T}})$.

Remark: (G,p) is independent if it has no nonzero self-stress.

$\to$ This implies that $\ker(\mathbf{R}(G,p)^{\mathsf{T}}) = \{0\}$, i.e. each rows of $\mathbf{R}(G,p)$ is linearly independent and hence (G,p) is independent.

Def 4.6) Strictly Convex Polyhedral framework

> A ***strictly convex polyhedral framework*** is a framework in $\mathbb{R}^3$ formed by
>
> 1. placing a joint at each vertex of a strictly convex polyhedron;
> 2. placing a bar along each edge of the polyhedron;
> 3. triangulating each convex face of the polyhedron and placing a bar along each edge of the triangulation.

Remark: For a strictly convex polyhedral framework, we have $|E(G)| = 3|V(G)| - 6$ (this follows from Euler's formula). Thus, such frameworks are statically rigid if and only if they do not have a nonzero self-stress.

Now, prove the Alexandrov's theorem. Before stating the theorem and proof, introduce an important lemma proposed by Cauchy.

Lem 4.7) Cauchy's Combinatorial lemma

> Let L be a finite graph on the two-sphere with no loops and no region bounded by just two edges of L.
>
> Mark the edges of L randomly with $+$ and $-$ and N_a be the number of sign changes as one circles around the vertex a.
>
> Let $N = \sum N_a$ be the total number of sign changes.

> Let V be the number of vertices of L. Then $N \leq 4V - 8$.

Thm 4.8) Alexandrov's theorem (weak version, 1940's)

> Every strictly convex polyhedral framework is statically rigid in 3-space and is independent.

Proof

First, **claim: Every strictly convex polyhedral framework is independent.**

Try induction on $|V(G)|$.

i) Base case: $|V(G)| \leq 3$

Trivially, if $|V(G)| \leq 3$, then a framework (G, p) cannot hold a self-stress, i.e. $\ker(\mathbf{R}(G,p))^\mathsf{T} = \{0\}$. But let's show them formally.

First, if $|V(G)| = 1$ (single joint without a bar), then obviously, such frameworks cannot hold self-stresses (there is no bar to assign a stress).

Next, if $|V(G)| = 2$ (a single bar with endpoints), then the rigidity matrix is of the form

$$\begin{aligned}\mathbf{R}(G,p) &= [p_1 - p_2 \quad p_2 - p_1] \\ &= [x_1 - x_2 \quad y_1 - y_2 \quad z_1 - z_2 \quad -(x_1 - x_2) \quad -(y_1 - y_2) \quad -(z_1 - z_2)] \in \mathbb{R}^{1\times 6}\end{aligned}$$

So, $\mathbf{R}(G,p)^\mathsf{T}\omega = \mathbf{0}$ and hence

$$\begin{bmatrix} x_1 - x_2 \\ y_1 - y_2 \\ z_1 - z_2 \\ x_2 - x_1 \\ y_2 - y_1 \\ z_2 - z_1 \end{bmatrix} \cdot \underbrace{\omega}_{\text{scalar}} = \begin{bmatrix} (x_1 - x_2)\cdot\omega \\ (y_1 - y_2)\cdot\omega \\ (z_1 - z_2)\cdot\omega \\ (x_2 - x_1)\cdot\omega \\ (y_2 - y_1)\cdot\omega \\ (z_2 - z_1)\cdot\omega \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \in \mathbb{R}^{6\times 1}$$

Thus, $\omega = 0$ and hence, $\ker(\mathbf{R}(G,p)^\mathsf{T}) = \{0\}$.

Finally, if $|V(G)| = 3$, then since we assumed a strictly convex polyhedral framework, (G, p) be a noncolinear triangle.

Similarly,

$$\mathbf{R}(G,p) = \begin{bmatrix} p_1 - p_2 & p_2 - p_1 & 0 \\ 0 & p_2 - p_3 & p_3 - p_2 \\ p_1 - p_3 & 0 & p_3 - p_1 \end{bmatrix} \in \mathbb{R}^{3\times 9} \qquad \text{and} \qquad \omega = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \in \mathbb{R}^{3\times 1}$$

and each ω_i must be zero for all $i = 1, 2, 3$.

ii) Inductive hypothesis

Assume that any subframework containing k or fewer joints (i.e. $|V(G)| \leq k$) cannot hold a self-stress ($\ker(\mathbf{R}(G,p)^\mathsf{T}) = \{\mathbf{0}\}$).

iii) Inductive step

Consider a subframework H with $k+1$ joints and assume it has a nontrivial self-stress ω_{ij} for each pair (i, j) where $1 \leq i < j \leq n$. (ω_{ij} denotes the stress assigned to the bar joining joint i and joint j)

Now, try to eliminate any bars where $\omega_{ij} = 0$.

If removing these zero-stress bars leave a joint completely isolated, then such a joint can also be removed.

However, if there exists such a joint (every incident bar doesn't hold a stress and hence isolated), say v_0, then $H - v_0$ be a subframework of G with nontrivial self-stress having only k joints.

This contradicts the assumption and hence we can conclude that every remaining joint is not isolated, i.e. connected to at least one bar with nonzero stress.

Let a_0 be a joint of H after removing all bars with zero stresses. Then, for a_0 to remain in equilibrium, the forces exerted by its connected bars must perfectly balance.

As we only consider a strictly convex polyhedral framework, the remaining bars are located in shape of umbrella.

The sequence of these forces will alternate between pushes and pulls as you evaluate the bars circularly around a_0.

The total number of sign changes around the joint a_0 must be an even number.

Case 1) Zero sign changes (ω_{0j} have same sign for all j)

WLOG, let $\omega_{0j} > 0$ for all j.

Since the polyhedron is strictly convex, there is a plane supporting such a polyhedron at a_0 that places all other connected joints into a single open half-space.

Let n be a normal vector of a plane. Then, since $n \cdot (a_0 - a_j) > 0$ for all j,

$$n \cdot \left(\sum_{j \in \mathcal{I}} \omega_{0j}(a_0 - a_j) \right) = \sum_{j \in \mathcal{I}} \omega_{0j}(n \cdot (a_0 - a_j)) > 0$$

where $\mathcal{I}$ is an index set of neighborhoods of a_0.

So, it is impossible to make $\sum_j \omega_{0j}(a_0 - a_j) = 0$.

Thus, case 1 never occurs.

Case 2) Two sign changes

In this case, a unique plane can be defined by a_0 and two specific bars that form the boundary between two groups. (Two bars uniquely define a plane)

WLOG, let n towards the side where the bars with positive stress.

Due to the strict convexity of polyhedron, the dot products of the remaining bars to n will perfectly align with their stress signs, i.e.

Since n towards the side where the bars with $\omega_{0j} > 0$, we have

$$n \cdot \left(\sum_{j \in \mathcal{I}^+} \omega_{0j}(a_0 - a_j) \right) = \sum_{j \in \mathcal{I}^+} \omega_{0j}(n \cdot (a_0 - a_j)) > 0$$

Again, since n towards the opposite side where the bars with $\omega_{0j} < 0$, we have

$$n \cdot \left(\sum_{j \in \mathcal{I}^-} \omega_{0j}(a_0 - a_j) \right) = \sum_{j \in \mathcal{I}^-} \omega_{0j}(n \cdot (a_0 - a_j)) > 0.$$

where $\mathcal{I}^+$ and $\mathcal{I}^-$ denote the index sets of joints where $\omega_{0j} > 0$ and $\omega_{0j} < 0$, respectively.

Thus, every joint forces a_0 to move toward n and hence, it is impossible for a_0 to remain equilibrium.

Case 3) More than two sign changes

Since the number of sign changes must be even, there are at least four sign changes.
Then, $N = \sum N_a$ must be larger than $4V$, i.e.

$$N \geq 4V$$

However, by the Cauchy's combinatorial lemma, $N \leq 4V - 8$ and hence, contradicts.

Therefore, we've observed that every case cannot occur and hence we can conclude that the assumption ($\exists$ nontrivial ω_{ij}) was wrong.

So, $\ker(\mathbf{R}(G,p)^\mathsf{T}) = \{\mathbf{0}\}$, i.e. (G,p) is independent.

Next, **claim: Independency implies Statical rigidity**

Since $\ker(\mathbf{R}(G,p)^\mathsf{T}) = \{\mathbf{0}\}$, $\text{nullity}(\mathbf{R}(G,p)^\mathsf{T}) = 0$.

By rank-nullity theorem, $\mathbf{R}(G,p)^\mathsf{T}$ has full column rank and automatically, $\mathbf{R}(G,p)$ has full row rank.

$$\text{rank}(\mathbf{R}(G,p)^\mathsf{T}) = \text{rank}(\mathbf{R}(G,p)) = |E(G)|$$

By the Euler's formula, for a strict(triangulated) convex polyhedron, $E = 3V - 6$ and hence

$$\text{rank}(\mathbf{R}(G,p)) = 3V - 6$$

Here, since $d = 3$ and $n = V$,

$$dn - \binom{d+1}{2} = 3V - 6$$

and hence by the theorem 2.17, (G,p) is infinitesimally rigid. Finally, since infinitesimal rigidity is equivalent to the static rigidity, (G,p) is statically rigid as well.

Hence, proved. $\square$

This theorem further generalized.

Thm 4.9) Alexandrov's theorem (1940's)

> If a strictly convex polyhedral framework is modified by
>
> 1. adding joints along natrural edges
> 2. replacement of the subframework on each face by a triagulation of the face polygon with the added joints

> then the modified framework is statically rigid in 3-space.

Alexandrov's theorem can be generalized to an arbitrary dimension:

Theorem 4.10) Extension of Alexandrov's theorem into higher dimension

> If a strictly convex d-polytopial framework $(d \geq 3)$ is formed in d-space by
>
> 1. the placement of a joint at each natural vertex;
> 2. the replacement of each 2-face by a subframework which triangulates the polygon
>
> then this framework is statically rigid in d-space.